Information in Noise: Strategic Trading under Autocorrelated Uninformed Orders

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Abstract

This paper develops a strategic trading model in which uninformed orders are allowed to exhibit a general correlation structure that generates autocorrelation in the order flow. Since the order flow is predictable, informed traders and the market maker not only need to infer information about the asset value, but also forecast future order flows. The correlation structure of uninformed orders has significant effects on trading strategies, market liquidity, and the informativeness of prices. Since the empirical autocorrelation in order flows is likely to come from uninformed traders, strategic trading models should not assume them to be simply noise.

Keywords: Kyle model, autocorrelated order flow, strategic trading, asymmetric information, signal processing, price impact, market microstructure.

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Strategic trading models usually classify traders as being either informed or uninformed.[1] The latter are often assumed to be noise traders who submit orders completely randomly due to liquidity or rebalancing needs. However, this is a rather strong assumption. Just because most traders do not possess non-public information does not imply that their trades should follow a random walk. So what if orders submitted by uninformed traders are not simply noise? What if instead there is a pattern to uninformed orders? This paper develops a strategic trading model that allows uninformed orders to exhibit a general correlation structure which generates autocorrelation in the order flow.

The first strategic trading model was developed by Kyle (1985) and ever since its original publication the model has inspired a large body of literature.[2] A key result in strategic trading models with long-lived private information is that informed traders place their orders in such a way that their trades are unpredictable to the market. That is, due to the trading strategy of informed traders, the market cannot infer whether informed traders are going to submit buy or sell orders. Since orders by uninformed traders are usually assumed to be white noise, the aggregate order flow is uncorrelated over time in these models.

However, as studies by Hasbrouck (1991a,b) and Brogaard, Hendershott, and Riordan (2019) show, there is autocorrelation in empirical order flows. Consequently, if informed traders have an incentive to be unpredictable with their trades, then the empirical autocorrelation should come from uninformed traders. Consider momentum strategies as a

[1] The existence of uninformed traders is a necessary condition for informed traders to participate in the market. See Kyle (1985), Glosten and Milgrom (1985), and Easley and O'Hara (1987).

[2] Extensions with multiple informed traders include Holden and Subrahmanyam (1992, 1994), Foster and Viswanathan (1994, 1996), Bernhardt and Miao (2004), Dridi and Germain (2009), Colla and Mele (2010), Ostrovsky (2012), Rostek and Weretka (2012), Lambert, Ostrovsky, and Panov (2018), and Sastry and Thompson (2019). Back (1992), Back, Cao, and Willard (2000), Back and Baruch (2004), and Collin-Dufresne and Fos (2016) extend the Kyle (1985) model in continuous time. Admati and Pfleiderer (1988), Kyle (1989), Seppi (1990), Foster and Viswanathan (1990), Degryse, Jong, and Kervel (2014), and Choi, Larsen, and Seppi (2019) incorporate strategic uninformed traders in their models. Caballe and Krishnan (1994) and Pasquariello (2007) study a multi-asset version of the one-period Kyle (1985) model with multiple informed traders.

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simple example. If a single stock performed strongly in the past, a momentum strategy would imply to buy this stock. Thus, if several uninformed traders were to follow this strategy, not only would the order flow be autocorrelated, but it might also be correlated with the true asset value and even non-public signals.

Moreover, there are also economic reasons why the autocorrelation in the order flow is likely to come from uninformed traders. In Degryse, Jong, and Kervel (2014) and Choi, Larsen, and Seppi (2019) autocorrelation in the order flow arises endogenously due to the existence of an uninformed trader who strategically rebalances his position to maintain a target level.[3] Since this uninformed trader must reach his target level before the end of the trading day, his orders are autocorrelated as a result. To conclude, it stands to reason that orders by uninformed traders generate autocorrelation in the order flow and that their orders might even be correlated with other variables of interest.

This paper contributes to the literature by allowing uninformed orders to exhibit cross-sectional correlation with the asset value and informed traders' signals as well as being autocorrelated over time. As a result, the aggregate order flow will also be autocorrelated and thus predictable. Even though these traders will be referred to as uninformed throughout this paper, one might in fact think of them as being partially informed traders who do not behave strategically. Being uninformed simply means not possessing relevant non-public information. This paper wants to ask the following questions: how should heterogeneously informed traders adjust their trading strategies under (auto)correlated uninformed orders? How does the (auto)correlation affect the value of non-public information? What is the impact on information efficiency in markets? How sensitive are prices to arriving orders when there is autocorrelation in the order flow?

This paper builds on the model by Foster and Viswanathan (1996) who were the first to

[3]Choi, Larsen, and Seppi (2019) also allow the uninformed trader to be partially informed. However, the model still generates autocorrelation in the order flow if the uninformed trader is fully uninformed.

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derive a dynamic trading equilibrium with heterogeneously informed traders.[4] As is usual in strategic trading models, it is assumed that the market consists of informed traders, a mass of uninformed traders, and a competitive market maker. Before trading starts, each informed trader receives a private signal about the asset's true value which leads to a "forecasting the forecasts of others" problem. A key result in Foster and Viswanathan (1996) is that the average signal of informed traders is a sufficient statistic to forecast the asset value. However, this is no longer true once uninformed orders are allowed to be (auto)correlated. The reason for this is that the average signal no longer captures all available information in the market since uninformed orders also contain information about the asset value.

A key result of this paper's model is that autocorrelation in the uninformed order flow implies uninformed orders to be conditionally correlated with the asset value and informed traders' signals in the cross-section. Thus, even if autocorrelated uninformed orders do not initially contain information about the asset value and non-public signals, they do so upon conditioning. Another main result is that only the innovation in the aggregate order flow has a price impact, implying that any form of predictability in the market gets priced immediately.

An extensive numerical analysis shows that (auto)correlated uninformed orders have significant effects on informed traders' expected profits and price discovery. Informed traders make the highest expected profits and prices are least revealing if uninformed orders are negatively correlated with the asset value in the cross-section. In this scenario private information is most valuable because uninformed traders effectively conceal informed traders' orders by pushing the price away from the true asset value. In contrast, informed traders' expected profits are lowest and informativeness of prices is highest if

[4]Back, Cao, and Willard (2000) study the Foster and Viswanathan (1996) model in continuous time. Other papers with heterogeneously informed traders include Foster and Viswanathan (1994), Bernhardt and Miao (2004), Dridi and Germain (2009), Colla and Mele (2010), Rostek and Weretka (2012), Ostrovsky (2012), and Lambert, Ostrovsky, and Panov (2018).

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the cross-sectional correlation between uninformed orders and the asset value is positive. In this scenario uninformed orders reveal information about the asset's true value, which generates competition for informed traders.

Moreover, the speed at which information gets incorporated into prices depends on the cross-sectional correlation between uninformed orders and informed traders' signals. Positive correlation implies that most information about the asset value gets revealed in the first periods, whereas more information gets incorporated into prices in later periods if the correlation is negative.

Finally, the very shape of the market maker's liquidity parameter is determined by the correlation structure of uninformed orders. In particular, the market maker's price sensitivity can be monotonically decreasing/increasing, constant, U-shaped, and even have a cubic form. Thus, (auto)correlated uninformed orders can explain various dynamics of market depth.

This research is related to Aase, Bjuland, and Øksendal (2012) and Lambert, Ostrovsky, and Panov (2018). Aase, Bjuland, and Øksendal (2012) consider a model with one perfectly informed trader and allow the uninformed order flow to be correlated with the asset value. The authors find a negative relationship between the informed trader's profits and the correlation between uninformed orders and the asset value. Additionally, informativeness of prices is at its lowest if uninformed orders are negatively correlated with the asset value.

In contrast, Lambert, Ostrovsky, and Panov (2018) consider multiple informed traders with heterogeneous information and allow for a most general correlation structure between the asset value, informed traders' signals, and the uninformed order flow.[5] They also find that correlated uninformed orders have significant effects on informed traders' profits and price discovery. Moreover, the authors provide general results on equilibrium existence

[5]Lambert, Ostrovsky, and Panov (2018) also allow the market maker to receive a signal that is correlated with the asset value, informed traders' signals, and the uninformed order flow.

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and uniqueness and derive general closed-form equilibrium characterizations.

However, both papers focus on a one-period model, whereas this paper wants to study a dynamic trading game. Even though it will be necessary to impose symmetry conditions for (co)variances in order to keep the analysis tractable in the dynamic setting, this paper can be viewed as a multi-period extension of the one-period model in Lambert, Ostrovsky, and Panov (2018).

The rest of this paper is organized as follows. Section I explains the general model framework and Section II describes the conjectured equilibrium. In Section III the updating process of informed traders along equilibrium and off-equilibrium paths is analyzed. Moreover, necessary and sufficient conditions for a dynamic trading equilibrium are provided. Finally, Section IV numerically evaluates the model and Section V concludes.

I. The Model

Let the trading day be standardized to the interval $[0, 1]$. Trading takes place over $N \in \mathbb{N}$ sequential batch auctions at equidistant time points $1/N$. The market consists of $M \in \mathbb{N}$ risk-neutral informed traders, a mass of uninformed traders, and one risk-neutral competitive market maker, who all trade for a single risky asset with stochastic value v and variance $\Sigma_0^v \equiv \mathrm{Var}[v]$. Trading takes place as follows:

(i) In each period $n \in \{1, \ldots, N\}$ informed and uninformed traders place their orders to the market maker.

(ii) The market maker observes the aggregate order flow and sets the clearing price.

(iii) The clearing price and aggregate order flow become public information.

(iv) Steps (i) to (iii) are repeated until the last trading period N is reached.

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(v) After the last round of trading the liquidation value of the asset v gets publicly announced and profits of the asset holders are realized.

Before the start of the trading game each informed trader $i \in \{1, \ldots, M\}$ receives a private signal $s_{i,0}$ about the asset's true value. Denote the signal vector as $\boldsymbol{s} \equiv [s_{1,0}, \ldots, s_{M,0}]'$ and let $\Sigma_0^{\boldsymbol{s}}$ be the corresponding covariance matrix with diagonals $\Sigma_0^{s_{i,0}} \equiv \text{Var}[s_{i,0}]$ and off-diagonals $\Sigma_0^{s_{i,0},s_{j,0}} \equiv \text{Cov}[s_{i,0}, s_{j,0}]$ where $j \in \{1, \ldots, M\}$ and $j \neq i$. Moreover, define the covariance between the asset value and informed trader i's signal as $\Sigma_0^{v,s_{i,0}} \equiv \text{Cov}[v, s_{i,0}]$. To keep future analysis tractable, the following symmetry conditions are imposed for all $i, j, k \in \{1, \ldots, M\}$:

$$\Sigma_0^{s_{i,0}} = \Sigma_0^{s_{j,0}}, \quad \Sigma_0^{s_{i,0},s_{j,0}} = \Sigma_0^{s_{i,0},s_{k,0}}, \quad \Sigma_0^{s_{i,0},s_{j,0}} = \Sigma_0^{s_{k,0},s_{j,0}}, \quad \Sigma_0^{v,s_{i,0}} = \Sigma_0^{v,s_{j,0}}.$$

The symmetry in (co)variances is a crucial assumption for the analysis that follows as it will keep the dimensionality space small.

Next, in each period $n \in \{1, \ldots, N\}$ uninformed traders submit the aggregate order u_n. Collect these orders into the vector $\boldsymbol{u} \equiv [u_1, \ldots, u_N]'$ and let $\Sigma_0^{\boldsymbol{u}}$ be the corresponding covariance matrix. Instead of assuming uninformed orders to be simply noise, the model will allow them to be correlated with the asset value and informed traders' signals in the cross-section as well as being autocorrelated over time.

To this end, assume that $\Sigma_0^{\boldsymbol{u}}$ has diagonals $\Sigma_0^{u_n} \equiv \text{Var}[u_n]$ and off-diagonals $\Sigma_0^{u_n,u_m} \equiv \text{Cov}[u_n, u_m]$ where $m \in \{1, \ldots, N\}$ and $m \neq n$. Moreover, define the covariance between the asset value and the uninformed order flow in period n as $\Sigma_0^{v,u_n} \equiv \text{Cov}[v, u_n]$ and the covariance between informed trader i's signal and the uninformed order flow in period n as $\Sigma_0^{s_{i,0},u_n} \equiv \text{Cov}[s_{i,0}, u_n]$. In addition, to keep future analysis tractable, the following symmetry conditions are imposed for all $n, m, l \in \{1, \ldots, N\}$ and all $i, j \in \{1, \ldots, M\}$:

$$\Sigma_0^{u_n} = \Sigma_0^{u_m}, \quad \Sigma_0^{u_n,u_m} = \Sigma_0^{u_n,u_l}, \quad \Sigma_0^{u_n,u_m} = \Sigma_0^{u_l,u_m}, \quad \Sigma_0^{v,u_n} = \Sigma_0^{v,u_m}, \quad \Sigma_0^{s_{i,0},u_n} = \Sigma_0^{s_{j,0},u_m}.$$

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Allowing for (auto)correlation in the uninformed order flow is the main contribution of this paper. As a result, the aggregate order flow will also be autocorrelated in the model. At first glance, the uninformed order vector $\boldsymbol{u}$ could be interpreted as a signal vector just like $\boldsymbol{s}$. However, the crucial difference is that signals are realized simultaneously before the beginning of the first trading period, whereas uninformed orders are only realized sequentially over time. Consequently, informed traders will not only need to infer information about other informed traders' signals, but also forecast future uninformed order flows.

Let $\Sigma_0^{v,\boldsymbol{s}}$ be the $(1 \times M)$ vector with entries $\Sigma_0^{v,s_{i,0}}$, $\Sigma_0^{v,\boldsymbol{u}}$ be the $(1 \times N)$ vector with entries Σ_0^{v,u_n}, and $\Sigma_0^{\boldsymbol{s},\boldsymbol{u}}$ be the $(M \times N)$ matrix with entries $\Sigma_0^{s_{i,0},u_n}$. It is assumed that the asset value, informed traders' signals, and uninformed orders are jointly normally distributed as follows:

$$\begin{pmatrix} v \\ \boldsymbol{s} \\ \boldsymbol{u} \end{pmatrix} \sim \mathcal{N}\left[\begin{pmatrix} 0 \\ \vdots \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} \underset{(1\times1)}{\Sigma_0^{v}} & \underset{(1\times M)}{\Sigma_0^{v,\boldsymbol{s}}} & \underset{(1\times N)}{\Sigma_0^{v,\boldsymbol{u}}} \\ \underset{(M\times1)}{\Sigma_0^{\boldsymbol{s},v}} & \underset{(M\times M)}{\Sigma_0^{\boldsymbol{s}}} & \underset{(M\times N)}{\Sigma_0^{\boldsymbol{s},\boldsymbol{u}}} \\ \underset{(N\times1)}{\Sigma_0^{\boldsymbol{u},v}} & \underset{(N\times M)}{\Sigma_0^{\boldsymbol{u},\boldsymbol{s}}} & \underset{(N\times N)}{\Sigma_0^{\boldsymbol{u}}} \end{pmatrix} \right]. \tag{1}$$

Denote the covariance matrix in (1) as $\boldsymbol{\Sigma_0}$ and assume the distribution is common knowledge in the market.[6]

Finally, there is a risk-neutral competitive market maker who observes the aggregate order flow and sets the clearing price. Suppose each informed trader $i \in \{1, \ldots, M\}$ submits a quantity $x_{i,n}$ in period $n \in \{1, \ldots, N\}$, then the aggregate order flow y_n in period n is defined as

$$y_n \equiv \sum_{i=1}^{M} x_{i,n} + u_n. \tag{2}$$

[6]Naturally, restrictions are required to ensure that the covariance matrix $\boldsymbol{\Sigma_0}$ in (1) is positive definite. Numerically, this will put upper bounds on the absolute values of covariances.

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The market maker cannot distinguish whether a single order is informed or uninformed and perfect competition implies that the clearing price p_n in period n is set to her conditional expectation of the asset value:

$$p_n \equiv \mathbb{E}\left[v \middle| y_1, \ldots, y_n\right], \tag{3}$$

where the order flow history $(y_1, \ldots, y_n)$ is the available information to the market maker after n periods of trading.

Consider the objective function of any informed trader $i \in \{1, \ldots, M\}$. In each trading period $n \in \{1, \ldots, N\}$ he maximizes his terminal conditional expected profit. The Bellman equation for his dynamic programming problem writes

$$\max_{x_{i,n}} \mathbb{E}\left[(v - p_n)x_{i,n} + V_{i,n} \middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right], \tag{4}$$

where $V_{i,n}$ is informed trader i's value function after n rounds of trading. Naturally, the value function depends on state variables that will need to be determined throughout the paper. Moreover, note that informed trader i's information set in period n consists of his private signal $s_{i,0}$, the order flow history $(y_1, \ldots, y_{n-1})$, and his individual trading history $(\tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1})$.[7] Following Foster and Viswanathan (1996), an equilibrium of the dynamic trading game is defined as follows.

Definition 1. *The strategies* $\{x_{1,n}, \ldots, x_{M,n}, p_n\}_{n=1}^N$ *form a Bayesian Nash equilibrium of the dynamic trading game if the following two conditions hold:*

(i) Given the strategies $\{x_{1,n}, \ldots, x_{i-1,n}, x_{i+1,n}, \ldots, x_{M,n}, p_n\}_{n=1}^N$*, the strategy* $\{x_{i,n}\}_{n=1}^N$ *maximizes the objective function (4) for each informed trader* $i \in \{1, \ldots, M\}$ *in every period* $n \in \{1, \ldots, N\}$*.*

[7] The individual trading history is written with a "tilde" to stress that the history can be arbitrary. By definition, the Bellman equation accounts for both optimal and suboptimal play in the past.

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(ii) Given the strategies $\{x_{1,n}, \ldots, x_{M,n}\}_{n=1}^{N}$*, the market maker's pricing rule* $\{p_n\}_{n=1}^{N}$ *satisfies (3) in every period* $n \in \{1, \ldots, N\}$*.*

There are two important things to note about Definition 1. First, the Bellman equation (4) requires informed trader i's strategy $x_{i,n}$ in period n to be optimal given any arbitrary past trading history $(\tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1})$. Thus, one will need to distinguish between trading along the equilibrium path and trading along off-equilibrium paths. Second, deviations by any informed trader $i \in \{1, \ldots, M\}$ from his optimal strategy are unobservable to other informed traders and the market maker. Consequently, there are no off-equilibrium beliefs in the model.

II. The Conjectured Equilibrium

Following Kyle (1985) and its extensions, this paper wants to derive a linear trading equilibrium. To this end, conjecture that along the equilibrium path informed traders' optimal strategies will be of the same form as in Foster and Viswanathan (1996), that is, for all $i \in \{1, \ldots, M\}$ and all $n \in \{1, \ldots, N\}$:

$$x_{i,n} = \beta_n s_{i,n-1}, \tag{5}$$

where $s_{i,n-1} \equiv s_{i,0} - \mathbb{E}\left[s_{i,0} \middle| y_1, \ldots, y_{n-1}\right]$ is the part of informed trader i's private information that is not known to the market maker after $n-1$ periods of trading. Thus, conjecture that even when uninformed orders are (auto)correlated, informed traders will trade in such a manner that their orders are unpredictable from the market maker's point of view. However, it is important to stress that (5) only characterizes the strategy that will be played along the equilibrium path and does not yet account for off-equilibrium play.

Although informed traders' orders are unpredictable, the aggregate order flow will be

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autocorrelated since

$$\mathbb{E}\left[y_n \middle| y_1, \ldots, y_{n-1}\right] = \mathbb{E}\left[\sum_{i=1}^{M} \beta_n s_{i,n-1} + u_n \middle| y_1, \ldots, y_{n-1}\right] = \mathbb{E}\left[u_n \middle| y_1, \ldots, y_{n-1}\right], \tag{6}$$

that is, the autocorrelation in the aggregate order flow comes solely from uninformed traders. This has the following implications for the market maker's updating processes along equilibrium play.[8]

Lemma 1. *Suppose informed trading strategies are as in (5). Then for all* $n \in \{1, \ldots, N\}$, $i \in \{1, \ldots, M\}$, *and* $t \in \{1, \ldots, N-n\}$ *the market maker recursively updates her beliefs as follows:*

$$p_n \equiv \mathbb{E}\left[v \middle| y_1, \ldots, y_n\right] = p_{n-1} + \lambda_n (y_n - r_{n,n-1}), \tag{7}$$

$$t_{i,n} \equiv \mathbb{E}\left[s_{i,0} \middle| y_1, \ldots, y_n\right] = t_{i,n-1} + \zeta_n (y_n - r_{n,n-1}), \tag{8}$$

$$r_{n+t,n} \equiv \mathbb{E}\left[u_{n+t} \middle| y_1, \ldots, y_n\right] = r_{n,n-1} + \theta_n (y_n - r_{n,n-1}), \tag{9}$$

where λ_n, ζ_n, *and* θ_n *are projection coefficients.*

Proof. See Appendix A.

There are two important things to note about Lemma 1. First, prices and beliefs are linear functions of $y_n - r_{n,n-1}$, that is, the innovation in the aggregate order flow. Thus, this paper confirms the results by Admati and Pfleiderer (1988), Degryse, Jong, and Kervel (2014), and Choi, Larsen, and Seppi (2019) that the predictable part of the aggregate order flow, in this case $r_{n,n-1}$, has no price impact. This implies that any form of predictability in the market gets incorporated into prices and beliefs immediately.

Second, in contrast to Foster and Viswanathan (1996), the market maker needs to forecast future uninformed orders, which according to (6) is equivalent to forecasting

[8] The market maker has no off-equilibrium beliefs since off-equilibrium play is unobservable.

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future aggregate order flows. It can be inferred from (9) that $r_{n+t,n}$ is independent of the forecasting horizon t. Consequently, it is sufficient to forecast uninformed orders only one period ahead, that is, after n periods $r_{n+1,n}$ is a sufficient statistic for forecasting all future uninformed orders. This is an important property since it keeps the dimensionality space small. Otherwise, the market maker would need to forecast each future uninformed order flow separately, resulting in a dimensionality problem with a growing number of trading periods.

While the market maker is also able to smooth her beliefs about past uninformed orders, it is not necessary for solving the model. More importantly, the smoothing process is different for each past uninformed order flow, that is, for all $n \in \{1, \ldots, N\}$ one can show that $r_{m,n} \neq r_{l,n}$ for all $m, l \in \{1, \ldots, n\}$ and $m \neq l$. Thus, to avoid the curse of dimensionality, this paper does not keep track of the smoothing process.[9]

It will prove helpful to consider uncertainty from the market maker's point of view. To this end, define for all $n \in \{1, \ldots, N\}$ and all $i, j \in \{1, \ldots, M\}$ the following conditional (co)variances:

$$\begin{aligned}
\Sigma_n^{v} &\equiv \operatorname{Var}\left[v \middle| y_1, \ldots, y_n\right], & \Sigma_n^{s_{i,0}} &\equiv \operatorname{Var}\left[s_{i,0} \middle| y_1, \ldots, y_n\right], \\
\Sigma_n^{u_{n+1}} &\equiv \operatorname{Var}\left[u_{n+1} \middle| y_1, \ldots, y_n\right], & \Sigma_n^{v,s_{i,0}} &\equiv \operatorname{Cov}\left[v, s_{i,0} \middle| y_1, \ldots, y_n\right], \\
\Sigma_n^{v,u_{n+1}} &\equiv \operatorname{Cov}\left[v, u_{n+1} \middle| y_1, \ldots, y_n\right], & \Sigma_n^{s_{i,0},s_{j,0}} &\equiv \operatorname{Cov}\left[s_{i,0}, s_{j,0} \middle| y_1, \ldots, y_n\right], \\
\Sigma_n^{s_{i,0},u_{n+1}} &\equiv \operatorname{Cov}\left[s_{i,0}, u_{n+1} \middle| y_1, \ldots, y_n\right], & \Sigma_n^{u_{n+1},u_{n+2}} &\equiv \operatorname{Cov}\left[u_{n+1}, u_{n+2} \middle| y_1, \ldots, y_n\right].
\end{aligned}$$

These (co)variances are conditional on the market maker's information after n rounds of trading, which consists of the aggregate order flow history $(y_1, \ldots, y_n)$. In particular, Σ_n^v measures how much uncertainty about the asset value remains after n trading periods.

[9] One can show that $\mathbb{E}\left[v \middle| \boldsymbol{s}, \boldsymbol{u}\right] = \mu \sum_{i=1}^{M} s_{i,0} + \omega \sum_{n=1}^{N} u_n \equiv \hat{v}$, where μ and ω are constants. While $\hat{v}$ is a sufficient statistic for solving the model, it also leads to a dimensionality issue since one would need to smooth past uninformed orders.

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The following lemma shows how these conditional (co)variances are related to each other.

Lemma 2. *Suppose informed trading strategies are as in (5). Then for all* $n \in \{1, \ldots, N\}$ *and* $i, j \in \{1, \ldots, M\}$ *it holds that*

$$\Sigma_{n-1}^{s_{i,0}} - \Sigma_n^{s_{i,0}} = \Sigma_{n-1}^{s_{i,0}, s_{j,0}} - \Sigma_n^{s_{i,0}, s_{j,0}}, \tag{10}$$

$$\Sigma_{n-1}^{u_n} - \Sigma_n^{u_{n+1}} = \Sigma_{n-1}^{u_n, u_{n+1}} - \Sigma_n^{u_{n+1}, u_{n+2}}. \tag{11}$$

Moreover, the market maker's liquidity parameter λ_n *in (7) satisfies the following relation:*

$$\lambda_n = \begin{pmatrix} \Sigma_{n-1}^{v,\boldsymbol{s}} & \Sigma_{n-1}^{v,u_n} \end{pmatrix} \begin{pmatrix} \Sigma_{n-1}^{\boldsymbol{s}} & \Sigma_{n-1}^{\boldsymbol{s},u_n} \\ \Sigma_{n-1}^{u_n,\boldsymbol{s}} & \Sigma_{n-1}^{u_n} \end{pmatrix}^{-1} \begin{pmatrix} \zeta_n \\ \vdots \\ \zeta_n \\ \tilde{\theta}_n \end{pmatrix}, \tag{12}$$

where $\Sigma_{n-1}^{\boldsymbol{s}}$ *is the* $(M \times M)$ *conditional covariance matrix with diagonals* $\Sigma_{n-1}^{s_{i,0}}$ *and off-diagonals* $\Sigma_{n-1}^{s_{i,0}, s_{j,0}}$, $\Sigma_{n-1}^{v,\boldsymbol{s}}$ *is the* $(1 \times M)$ *conditional covariance vector with entries* $\Sigma_{n-1}^{v, s_{i,0}}$, $\Sigma_{n-1}^{\boldsymbol{s},u_n}$ *is the* $(M \times 1)$ *conditional covariance vector with entries* $\Sigma_{n-1}^{s_{i,0},u_n}$, *and* $\tilde{\theta}_n \neq \theta_n$ *is a projection coefficient.*

Proof. See Appendix B.

The first part of Lemma 2 implies for all $n \in \{1, \ldots, N-1\}$:

$$\Sigma_{n-1}^{s_{i,0}} - \Sigma_{n-1}^{s_{i,0}, s_{j,0}} \equiv \chi_s, \tag{13}$$

$$\Sigma_{n-1}^{u_n} - \Sigma_{n-1}^{u_n, u_{n+1}} \equiv \chi_u, \tag{14}$$

where χ_s and χ_u are constants. Thus, for both informed traders' signals and future uninformed orders the difference between variances and covariances is independent of the trading period.

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The second part of Lemma 2 implies that the market maker's update of the asset value is not proportional to her update of informed traders' signals:

$$p_n - p_{n-1} \not\propto \sum_{i=1}^{M} (t_{i,n} - t_{i,n-1}). \tag{15}$$

A key result in Foster and Viswanathan (1996) is that in a market with heterogeneously informed traders and independent uninformed orders the market maker only needs to forecast the average signal of informed traders. However, this is no longer true once (auto)correlation in uninformed orders is introduced. The reason for this is that the average signal is no longer a sufficient statistic for all available information in the market since uninformed orders also contain relevant information. Thus, (auto)correlated uninformed orders disentangle the one-to-one relationship between forecasting the asset value and forecasting the private information of informed traders.

III. The Bayesian Updating Process

This section addresses the dimensionality issue of the dynamic trading game and shows how Bayesian beliefs of informed traders are updated over time. Definition 1 requires one to distinguish between updating along the equilibrium path and updating along off-equilibrium paths. Deriving the updating process will help identify the state variables and the corresponding value function of informed traders' dynamic programming problem.

A. Updating along the Equilibrium Path

Following Foster and Viswanathan (1996), it is first shown that for every informed trader $i \in \{1, \ldots, M\}$ the individual past trading history is redundant information along the equilibrium path. Indeed, from (5), (8), and (9) one infers that in every period

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$n \in \{1, \ldots, N\}$ informed trader i's optimal trading strategy $x_{i,n}$ can be written as

$$\begin{aligned} x_{i,n} &= \beta_n s_{i,n-1} = \beta_n (s_{i,0} - t_{i,n-1}) = \beta_n \left(s_{i,0} - \sum_{t=1}^{n-1} \zeta_t [y_t - r_{t,t-1}] \right) \\ &= \beta_n \left(s_{i,0} - \sum_{t=1}^{n-1} \zeta_t \left[y_t - \sum_{r=1}^{t-1} \left(\prod_{s=r+1}^{t-1} \left[1 - \theta_s \right] \right) \theta_r y_r \right] \right). \end{aligned} \tag{16}$$

Equation (16) implies that informed trader i's optimal trading strategy in period n is a function of his private signal $s_{i,0}$ and the past aggregate order flow history $(y_1, \ldots, y_{n-1})$. Therefore, the individual trading history $(x_{i,1}, \ldots, x_{i,n-1})$ is redundant and the meaningful information for informed trader i is just $(s_{i,0}, y_1, \ldots, y_{n-1})$. This also implies that the only relevant information that informed trader i has relative to the market maker is his private signal $s_{i,0}$, which helps identify the state variable for updating along the equilibrium path.

Lemma 3. *Assume the market maker sets prices and forms beliefs as in (7), (8), and (9) and all informed traders $i, j \in \{1, \ldots, M\}$ where $i \neq j$ trade as in (5). Then $s_{i,n-1}$ is a sufficient statistic for informed trader i's updating processes after $n-1$ periods for all $n \in \{1, \ldots, N\}$. In particular, the updating processes are as follows:*

$$\mathbb{E}\left[v - p_{n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}, x_{i,1}, \ldots, x_{i,n-1} \right] = \eta_n s_{i,n-1}, \tag{17}$$

$$\mathbb{E}\left[s_{j,n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}, x_{i,1}, \ldots, x_{i,n-1} \right] = \phi_n s_{i,n-1}, \tag{18}$$

$$\mathbb{E}\left[u_n - r_{n,n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}, x_{i,1}, \ldots, x_{i,n-1} \right] = \psi_n s_{i,n-1}, \tag{19}$$

where η_n, ϕ_n, and ψ_n are projection coefficients.

Proof. See Appendix C.

Lemma 3 states that if all informed traders (including i) have played optimally in the first $n-1$ trading rounds, then the residual signal $s_{i,n-1} = s_{i,0} - t_{i,n-1}$ is a sufficient statistic for informed trader i to forecast the asset value, other informed traders' signals, and next

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period's uninformed order flow.[10] Thus, $s_{i,n-1}$ is the only state variable along equilibrium play. However, it is important to stress that these results only hold if informed trader i has played his equilibrium strategy in the first $n-1$ trading periods and are no longer valid along off-equilibrium paths. Additional state variables are required to account for off-equilibrium play.

B. Updating along Off-Equilibrium Paths

Without loss of generality, assume informed trader i has submitted an arbitrary order sequence $(\tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n})$ in the first $n \in \{1, \ldots, N\}$ trading periods while all other informed traders $j \in \{1, \ldots, M\}$ where $j \neq i$ follow the equilibrium strategy (5). Furthermore, suppose the market maker sets prices and forms beliefs as in (7), (8), and (9).

Since deviations from the equilibrium strategy are unobservable, informed trader i can manipulate other informed traders' and the market maker's beliefs about the asset value, signals, and uninformed orders. More importantly, if informed trader i manipulates beliefs by submitting arbitrary orders $(\tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n})$, the projection theorem for Gaussian random variables cannot be applied since $s_{i,n}$ and $(y_1, \ldots, y_n)$ need not be jointly normally distributed. In particular, $s_{i,n}$ will no longer be a sufficient forecasting statistic for informed trader i after n periods of trading.

To address this issue, closely follow Foster and Viswanathan (1996) and introduce the following notation for all $n \in \{1, \ldots, N\}$ and $j \in \{1, \ldots, M\}$:

$$\begin{aligned}
\hat{y}_n^i &\equiv \sum_{j=1}^{M} \beta_n \hat{s}_{j,n-1}^i + u_n, \\
\hat{s}_{j,n}^i &\equiv s_{j,0} - \hat{t}_{j,n}^i \text{ and } \hat{s}_{j,0}^i \equiv s_{j,0}, \\
\hat{p}_n^i &\equiv \hat{p}_{n-1}^i + \lambda_n \left(\hat{y}_n^i - \hat{r}_{n,n-1}^i \right) \text{ and } \hat{p}_0^i \equiv 0,
\end{aligned}$$

[10] Technically, informed traders need to form beliefs about all future uninformed order flows. However, as was shown in Lemma 1, it is sufficient to forecast uninformed orders only one period ahead.

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$$\hat{t}^i_{j,n} \equiv \hat{t}^i_{j,n-1} + \zeta_n \left(\hat{y}^i_n - \hat{r}^i_{n,n-1}\right) \text{ and } \hat{t}^i_{j,0} \equiv 0,$$
$$\hat{r}^i_{n+1,n} \equiv \hat{r}^i_{n,n-1} + \theta_n \left(\hat{y}^i_n - \hat{r}^i_{n,n-1}\right) \text{ and } \hat{r}^i_{1,0} \equiv 0.$$

The "hat" processes are the quantities that would have occurred along the equilibrium path in trading period n if informed trader i had followed his equilibrium strategy $\{\beta_m \hat{s}^i_{i,m-1}\}_{m=1}^n$. All results from the previous sections are applicable to these "hat" processes which denote the processes along the equilibrium path. With this notation in mind, an important result from Foster and Viswanathan (1996), that will help characterize off-equilibrium paths, is stated.

Lemma 4. *Assume the market maker sets prices and forms beliefs as in (7), (8), and (9) and all informed traders $j \in \{1, \ldots, M\}$ where $j \neq i$ trade as in (5). Suppose informed trader i has submitted an arbitrary sequence of orders $(\tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n})$ in the first $n \in \{1, \ldots, N\}$ trading periods. Then the following two information sets are equivalent:*

$$\left(s_{i,0}, y_1, \ldots, y_n, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n}\right) \equiv \left(s_{i,0}, \hat{y}^i_1, \ldots, \hat{y}^i_n, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n}\right). \tag{20}$$

Proof. See Appendix D.

Lemma 4 states that the equilibrium aggregate order flow history $(\hat{y}^i_1, \ldots, \hat{y}^i_n)$ is in informed trader i's information set even along off-equilibrium paths. Intuitively, informed trader i knows how his trades affect the beliefs and strategies of other informed traders and the market maker. Consequently, he can reconstruct the aggregate order flow history that would have occurred along the equilibrium path if he had played his equilibrium strategy. Since $(\hat{y}^i_1, \ldots, \hat{y}^i_n)$ is in informed trader i's information set, so is the history of the equilibrium quantities $\{\hat{s}^i_{i,m}, \hat{p}^i_m, \hat{t}^i_{j,m}, \hat{r}^i_{m+1,m}\}_{m=0}^n$.

Lemma 3 has shown that, along the equilibrium path, $\hat{s}^i_{i,n-1}$ is a sufficient forecasting statistic for informed trader i after $n-1$ periods of trading. The following lemma lists

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the additional state variables that are required for updating along off-equilibrium paths.

Lemma 5. *Assume the market maker sets prices and forms beliefs as in (7), (8), and (9) and all informed traders $j \in \{1, \ldots, M\}$ where $j \neq i$ trade as in (5). Suppose informed trader i has submitted an arbitrary sequence of orders $(\tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1})$ in the first $n-1$ trading rounds. Then $\hat{s}^i_{i,n-1}$, $\hat{p}^i_{n-1} - p_{n-1}$, $\hat{t}^i_{j,n-1} - t_{j,n-1}$, and $\hat{r}^i_{n,n-1} - r_{n,n-1}$ are sufficient statistics for informed trader i's updating processes after $n-1$ periods for all $n \in \{1, \ldots, N\}$. In particular, the updating processes are as follows:*

$$\mathbb{E}\left[v - p_{n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] = \eta_n \hat{s}^i_{i,n-1} + \left(\hat{p}^i_{n-1} - p_{n-1}\right), \tag{21}$$

$$\mathbb{E}\left[s_{j,n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] = \phi_n \hat{s}^i_{i,n-1} + \left(\hat{t}^i_{j,n-1} - t_{j,n-1}\right), \tag{22}$$

$$\mathbb{E}\left[u_n - r_{n,n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] = \psi_n \hat{s}^i_{i,n-1} + \left(\hat{r}^i_{n,n-1} - r_{n,n-1}\right). \tag{23}$$

Proof. See Appendix E.

In Foster and Viswanathan (1996) the residual signal $\hat{s}^i_{i,n-1}$ and the price deviation from equilibrium $\hat{p}^i_{n-1} - p_{n-1}$ are sufficient statistics for updating along off-equilibrium paths. However, this is no longer the case once one allows for (auto)correlation in the uninformed order flow. According to Lemma 5, the deviation in beliefs from equilibrium about informed traders' signals $\hat{t}^i_{j,n-1} - t_{j,n-1}$ and the deviation in beliefs from equilibrium about next period's uninformed order flow $\hat{r}^i_{n,n-1} - r_{n,n-1}$ are the additional state variables that are required for updating along off-equilibrium paths.

If uninformed orders are assumed to be white noise, the updating process of the asset value is proportional to the updating process of informed traders' signals. This is the reason why $\hat{t}^i_{j,n-1} - t_{j,n-1}$ is a redundant state variable in Foster and Viswanathan (1996). However, as was shown in (15), this proportionality breaks down if uninformed orders are allowed to exhibit a general correlation structure. Consequently, updates about the asset value have to be considered separately from updates about informed traders' signals

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since orders by uninformed traders also contain relevant information.

C. Necessary and Sufficient Conditions for Equilibrium

With the insights from Lemma 5 in mind, conjecture informed trader i's optimal strategy, that accounts for both equilibrium and off-equilibrium play, to be linear in the state variables. In particular, given any arbitrary order sequence $(\tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1})$, conjecture informed trader i's optimal strategy for the remaining trading periods $m \in \{n, \ldots, N\}$ to be of the form

$$x_{i,m} = \beta_m \hat{s}^i_{i,m-1} + \gamma_m \left(\hat{p}^i_{m-1} - p_{m-1}\right) + \alpha_m \left(\hat{t}^i_{j,m-1} - t_{j,m-1}\right) + \delta_m \left(\hat{r}^i_{m,m-1} - r_{m,m-1}\right). \quad (24)$$

Along the equilibrium path one has $p_{m-1} = \hat{p}^i_{m-1}$, $t_{j,m-1} = \hat{t}^i_{j,m-1}$, and $r_{m,m-1} = \hat{r}^i_{m,m-1}$ so that only $\beta_m \hat{s}^i_{i,m-1}$ remains. Thus, the originally conjectured optimal strategy (5), that will be played along the equilibrium path, is consistent with the strategy (24). Moreover, since informed traders are assumed to be risk-neutral, conjecture the value function $V_{i,n-1}$ of informed trader i after $n-1$ trading periods to be of quadratic form, that is, for all $i, j \in \{1, \ldots, M\}$ where $i \neq j$ and all $n \in \{1, \ldots, N\}$ conjecture:

$$\begin{aligned} V_{i,n-1} = {} & a_{n-1} \left(\hat{s}^i_{i,n-1}\right)^2 + b_{n-1} \left(\hat{p}^i_{n-1} - p_{n-1}\right)^2 + c_{n-1} \left(\hat{t}^i_{j,n-1} - t_{j,n-1}\right)^2 + d_{n-1} \left(\hat{r}^i_{n,n-1} - r_{n,n-1}\right)^2 \\ & + e_{n-1} \hat{s}^i_{i,n-1} \left(\hat{p}^i_{n-1} - p_{n-1}\right) + f_{n-1} \hat{s}^i_{i,n-1} \left(\hat{t}^i_{j,n-1} - t_{j,n-1}\right) + g_{n-1} \hat{s}^i_{i,n-1} \left(\hat{r}^i_{n,n-1} - r_{n,n-1}\right) \\ & + h_{n-1} \left(\hat{p}^i_{n-1} - p_n\right) \left(\hat{t}^i_{j,n-1} - t_{j,n-1}\right) + i_{n-1} \left(\hat{p}^i_{n-1} - p_{n-1}\right) \left(\hat{r}^i_{n,n-1} - r_{n,n-1}\right) \\ & + j_{n-1} \left(\hat{t}^i_{j,n-1} - t_{j,n-1}\right) \left(\hat{r}^i_{n,n-1} - r_{n,n-1}\right) + k_{n-1}. \end{aligned} \quad (25)$$

Along the equilibrium path only the first and last terms of the value function remain. However, since a Bayesian strategy also needs to be optimal along off-equilibrium paths, one needs to consider the additional terms. The following proposition states the main result of this paper.

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Proposition 1. *The strategies and beliefs (7), (8), (9), (21), (22), (23), and (24) form a symmetric linear Markov perfect equilibrium if for all $n \in \{1, \dots, N\}$ the following recursions hold:*

$$\beta_n = \frac{\eta_n - \lambda_n\psi_n - (1-\zeta_n\psi_n)(e_n\lambda_n + f_n\zeta_n + g_n\theta_n)}{\lambda_n[2+(M-1)\phi_n] - \zeta_n[1+(M-1)\phi_n](e_n\lambda_n + f_n\zeta_n + g_n\theta_n)},$$

$$\gamma_n = \frac{1 - 2b_n\lambda_n - h_n\zeta_n - i_n\theta_n}{\lambda_n(2 - 2b_n\lambda_n - h_n\zeta_n - i_n\theta_n) - \zeta_n(2c_n\zeta_n + h_n\lambda_n + j_n\theta_n) - \theta_n(2d_n\theta_n + i_n\lambda_n + j_n\zeta_n)},$$

$$\alpha_n = \frac{\begin{array}{c}(M-1)\beta_n\Big[\theta_n(2d_n\theta_n + i_n\lambda_n + j_n\zeta_n) - \lambda_n(1 - 2b_n\lambda_n - h_n\zeta_n - i_n\theta_n)\Big] \\ - [1-(M-1)\beta_n\zeta_n](2c_n\zeta_n + h_n\lambda_n + j_n\theta_n)\end{array}}{\lambda_n(2 - 2b_n\lambda_n - h_n\zeta_n - i_n\theta_n) - \zeta_n(2c_n\zeta_n + h_n\lambda_n + j_n\theta_n) - \theta_n(2d_n\theta_n + i_n\lambda_n + j_n\zeta_n)},$$

$$\delta_n = \frac{\zeta_n(2c_n\zeta_n + h_n\lambda_n + j_n\theta_n) - \lambda_n(1 - 2b_n\lambda_n - h_n\zeta_n - i_n\theta_n) - (1-\theta_n)(2d_n\theta_n + i_n\lambda_n + j_n\zeta_n)}{\lambda_n(2 - 2b_n\lambda_n - h_n\zeta_n - i_n\theta_n) - \zeta_n(2c_n\zeta_n + h_n\lambda_n + j_n\theta_n) - \theta_n(2d_n\theta_n + i_n\lambda_n + j_n\zeta_n)},$$

$$\lambda_n = \frac{M\beta_n\Sigma_{n-1}^{v,s_{i,0}} + \Sigma_{n-1}^{v,u_n}}{M\beta_n^2\Big[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\Big] + \Sigma_{n-1}^{u_n} + 2M\beta_n\Sigma_{n-1}^{s_{i,0},u_n}},$$

$$\zeta_n = \frac{\beta_n\Big[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\Big] + \Sigma_{n-1}^{s_{i,0},u_n}}{M\beta_n^2\Big[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\Big] + \Sigma_{n-1}^{u_n} + 2M\beta_n\Sigma_{n-1}^{s_{i,0},u_n}},$$

$$\theta_n = \frac{M\beta_n\Sigma_{n-1}^{s_{i,0},u_n} + \Sigma_{n-1}^{u_n,u_{n+1}}}{M\beta_n^2\Big[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\Big] + \Sigma_{n-1}^{u_n} + 2M\beta_n\Sigma_{n-1}^{s_{i,0},u_n}},$$

with value function coefficients

$$a_{n-1} = \beta_n\Big(\eta_n - \lambda_n\Big[\beta_n + (M-1)\beta_n\phi_n + \psi_n\Big]\Big) + a_n\Big([1-\zeta_n\beta_n]^2 - 2\zeta_n(1-\zeta_n\beta_n)\Big[\psi_n + (M-1)\beta_n\phi_n\Big]\Big),$$

$$b_{n-1} = b_n(1-\gamma_n\lambda_n)^2 + \gamma_n(1-\gamma_n\lambda_n)(1 - h_n\zeta_n - i_n\theta_n) + \gamma_n^2(c_n\zeta_n^2 + d_n\theta_n^2 + j_n\zeta_n\theta_n),$$

$$\begin{aligned} c_{n-1} = {} & c_n\Big(1 - \zeta_n\Big[\alpha_n + (M-1)\beta_n\Big]\Big)^2 + \Big[\alpha_n + (M-1)\beta_n\Big]^2(b_n\lambda_n^2 + d_n\theta_n^2 + i_n\lambda_n\theta_n) \\ & - \Big[\alpha_n + (M-1)\beta_n\Big]\Big[\alpha_n\lambda_n + \Big(1 - \zeta_n\Big[\alpha_n + (M-1)\beta_n\Big]\Big)(h_n\lambda_n + j_n\theta_n)\Big], \end{aligned}$$

$$\begin{aligned} d_{n-1} = {} & d_n\Big(1 - \theta_n[1+\delta_n]\Big)^2 + (1+\delta_n)^2(b_n\lambda_n^2 + c_n\zeta_n^2 + h_n\lambda_n\zeta_n) \\ & - (1+\delta_n)\Big[\delta_n\lambda_n + \Big(1 - \theta_n[1+\delta_n]\Big)(i_n\lambda_n + j_n\zeta_n)\Big], \end{aligned}$$

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$$
\begin{aligned}
e_{n-1} &= \Big[e_n(1-\gamma_n\lambda_n) - \gamma_n(f_n\zeta_n + g_n\theta_n)\Big]\Big(1-\zeta_n\Big[\beta_n + (M-1)\beta_n\phi_n + \psi_n\Big]\Big) \\
&\quad + \gamma_n\Big(\eta_n - \lambda_n\Big[\beta_n + (M-1)\beta_n\phi_n + \psi_n\Big]\Big) + \beta_n(1-\gamma_n\lambda_n), \\
f_{n-1} &= \Big[f_n\Big(1-\zeta_n\Big[\alpha_n + (M-1)\beta_n\Big]\Big) - \Big[\alpha_n + (M-1)\beta_n\Big](e_n\lambda_n + g_n\theta_n)\Big]\Big(1-\zeta_n\Big[\beta_n + (M-1)\beta_n\phi_n + \psi_n\Big]\Big) \\
&\quad + \alpha_n\Big(\eta_n - \lambda_n\Big[\beta_n + (M-1)\beta_n\phi_n + \psi_n\Big]\Big) - \lambda_n\beta_n\Big[\alpha_n + (M-1)\beta_n\Big], \\
g_{n-1} &= \Big[g_n\Big(1-\theta_n[1+\delta_n]\Big) - (1+\delta_n)(e_n\lambda_n + f_n\zeta_n)\Big]\Big(1-\zeta_n\Big[\beta_n + (M-1)\beta_n\phi_n + \psi_n\Big]\Big) \\
&\quad + \delta_n\Big(\eta_n - \lambda_n\Big[\beta_n + (M-1)\beta_n\phi_n + \psi_n\Big]\Big) - \lambda_n\beta_n(1+\delta_n), \\
h_{n-1} &= \Big[h_n(1-\gamma_n\lambda_n) - \gamma_n(2c_n\zeta_n + j_n\theta_n)\Big]\Big(1-\zeta_n\Big[\alpha_n + (M-1)\beta_n\Big]\Big) + \alpha_n(1-\gamma_n\lambda_n) \\
&\quad + \Big[\alpha_n + (M-1)\beta_n\Big]\Big[\gamma_n\Big[\theta_n(2d_n\theta_n + j_n\zeta_n) - \lambda_n(1-h_n\zeta_n - i_n\theta_n)\Big] - (1-\gamma_n\lambda_n)(2b_n\lambda_n + i_n\theta_n)\Big], \\
i_{n-1} &= \Big[i_n(1-\gamma_n\lambda_n) - \gamma_n(2d_n\theta_n + j_n\zeta_n)\Big]\Big(1-\theta_n[1+\delta_n]\Big) + \delta_n(1-\gamma_n\lambda_n) \\
&\quad + (1+\delta_n)\Big[\gamma_n\Big[\zeta_n(2c_n\zeta_n + j_n\theta_n) - \lambda_n(1-h_n\zeta_n - i_n\theta_n)\Big] - (1-\gamma_n\lambda_n)(2b_n\lambda_n + h_n\zeta_n)\Big], \\
j_{n-1} &= \Big[j_n\Big(1-\theta_n[1+\delta_n]\Big) - (1+\delta_n)(2c_n\zeta_n + h_n\lambda_n)\Big]\Big(1-\zeta_n\Big[\alpha_n + (M-1)\beta_n\Big]\Big) - \alpha_n\lambda_n(1+\delta_n) \\
&\quad + \Big[\alpha_n + (M-1)\beta_n\Big]\Big[(1+\delta_n)\Big[\lambda_n(2b_n\lambda_n + h_n\zeta_n) + \theta_n(i_n\lambda_n + j_n\zeta_n)\Big] - \lambda_n\delta_n \\
&\quad - \Big(1-\theta_n[1+\delta_n]\Big)(2d_n\theta_n + i_n\lambda_n)\Big], \\
k_{n-1} &= k_n + a_n\zeta_n^2\Big[\operatorname{Var}\Big[u_n - r_{n,n-1}\Big|s_{i,0}, y_1, \ldots, y_{n-1}\Big] + 2(M-1)\beta_n \operatorname{Cov}\Big[s_{j,n-1}, u_n - r_{n,n-1}\Big|s_{i,0}, y_1, \ldots, y_{n-1}\Big] \\
&\quad + (M-1)\beta_n^2\Big(\operatorname{Var}\Big[s_{j,n-1}\Big|s_{i,0}, y_1, \ldots, y_{n-1}\Big] + (M-2)\operatorname{Cov}\Big[s_{j,n-1}, s_{k,n-1}\Big|s_{i,0}, y_1, \ldots, y_{n-1}\Big]\Big)\Big],
\end{aligned}
$$

where $a_N = b_N = c_N = d_N = e_N = f_N = g_N = h_N = i_N = j_N = k_N = 0$ *and*

$$
\eta_n = \frac{\Sigma_{n-1}^{v,s_{i,0}}}{\Sigma_{n-1}^{s_{i,0}}},
$$

$$
\phi_n = \frac{\Sigma_{n-1}^{s_{i,0},s_{j,0}}}{\Sigma_{n-1}^{s_{i,0}}},
$$

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$$\psi_n = \frac{\Sigma_{n-1}^{s_{i,0},u_n}}{\Sigma_{n-1}^{s_{i,0}}}.$$

Also, for all $i, j, k \in \{1, \ldots, M\}$ where $i \neq j \neq k$:

$$\begin{aligned}
\text{Var}\left[u_n - r_{n,n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}\right] &= \Sigma_{n-1}^{u_n} - \psi_n^2 \Sigma_{n-1}^{s_{i,0}}, \\
\text{Var}\left[s_{j,n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}\right] &= (1-\phi_n^2)\Sigma_{n-1}^{s_{i,0}}, \\
\text{Cov}\left[s_{j,n-1}, s_{k,n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}\right] &= (1-\phi_n)\Sigma_{n-1}^{s_{i,0},s_{j,0}}, \\
\text{Cov}\left[s_{j,n-1}, u_n - r_{n,n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}\right] &= (1-\phi_n)\Sigma_{n-1}^{s_{i,0},u_n}.
\end{aligned}$$

In addition, the second order condition must hold in every period:

$$\lambda_n(2 - 2b_n\lambda_n - h_n\zeta_n - i_n\theta_n) - \zeta_n(2c_n\zeta_n + h_n\lambda_n + j_n\theta_n) - \theta_n(2d_n\theta_n + i_n\lambda_n + j_n\zeta_n) > 0.$$

Finally, the conditional (co)variances must satisfy the following recursions:

$$\begin{aligned}
\Sigma_n^{v} &= \Sigma_{n-1}^{v} - \lambda_n^2\Big(M\beta_n^2\Big[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\Big] + \Sigma_{n-1}^{u_n} + 2M\beta_n\Sigma_{n-1}^{s_{i,0},u_n}\Big), \\
\Sigma_n^{v,s_{i,0}} &= \Sigma_{n-1}^{v,s_{i,0}} - \lambda_n\zeta_n\Big(M\beta_n^2\Big[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\Big] + \Sigma_{n-1}^{u_n} + 2M\beta_n\Sigma_{n-1}^{s_{i,0},u_n}\Big), \\
\Sigma_n^{v,u_{n+1}} &= \Sigma_{n-1}^{v,u_n} - \lambda_n\theta_n\Big(M\beta_n^2\Big[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\Big] + \Sigma_{n-1}^{u_n} + 2M\beta_n\Sigma_{n-1}^{s_{i,0},u_n}\Big), \\
\Sigma_n^{s_{i,0}} &= \Sigma_{n-1}^{s_{i,0}} - \zeta_n^2\Big(M\beta_n^2\Big[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\Big] + \Sigma_{n-1}^{u_n} + 2M\beta_n\Sigma_{n-1}^{s_{i,0},u_n}\Big), \\
\Sigma_n^{s_{i,0},s_{j,0}} &= \Sigma_{n-1}^{s_{i,0},s_{j,0}} - \zeta_n^2\Big(M\beta_n^2\Big[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\Big] + \Sigma_{n-1}^{u_n} + 2M\beta_n\Sigma_{n-1}^{s_{i,0},u_n}\Big), \\
\Sigma_n^{s_{i,0},u_{n+1}} &= \Sigma_{n-1}^{s_{i,0},u_n} - \zeta_n\theta_n\Big(M\beta_n^2\Big[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\Big] + \Sigma_{n-1}^{u_n} + 2M\beta_n\Sigma_{n-1}^{s_{i,0},u_n}\Big), \\
\Sigma_n^{u_{n+1}} &= \Sigma_{n-1}^{u_n} - \theta_n^2\Big(M\beta_n^2\Big[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\Big] + \Sigma_{n-1}^{u_n} + 2M\beta_n\Sigma_{n-1}^{s_{i,0},u_n}\Big), \\
\Sigma_n^{u_{n+1},u_{n+2}} &= \Sigma_{n-1}^{u_n,u_{n+1}} - \theta_n^2\Big(M\beta_n^2\Big[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\Big] + \Sigma_{n-1}^{u_n} + 2M\beta_n\Sigma_{n-1}^{s_{i,0},u_n}\Big).
\end{aligned}$$

Proof. See Appendix F.

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Proposition 1 specifies the necessary and sufficient conditions for a dynamic trading equilibrium. As is the case with most discrete-time strategic trading models, including Foster and Viswanathan (1996), this paper does not have analytic expressions for the equilibrium so that Proposition 1 must be computed numerically. The corresponding algorithm is described in Appendix G.

While Σ_n^v is not technically required to solve the remaining system of equations, one still needs to keep track of it when it comes to the numerical analysis. Since an equilibrium is only sustainable if the market maker does not fully learn about the asset's true value before the final trading period, one has to ensure that $\Sigma_{N-1}^v > 0$.

Moreover, from the equations for θ_n, $\Sigma_n^{v,u_{n+1}}$, $\Sigma_n^{s_{i,0},u_{n+1}}$, and $\Sigma_n^{u_{n+1},u_{n+2}}$ it can be inferred that uninformed orders are conditionally correlated with the asset value and informed traders' signals in the cross-section if they are autocorrelated over time. Thus, even if autocorrelated uninformed orders do not initially contain information about the asset value and signals, they do so upon conditioning. Similarly, if uninformed orders are initially uncorrelated in the time series, they become conditionally autocorrelated as long as they initially contain information about signals.

Finally, note that the models by Kyle (1985), Holden and Subrahmanyam (1992), and Foster and Viswanathan (1996) are all nested in Proposition 1. If one assumes uninformed orders to be white noise, one obtains the Foster and Viswanathan (1996) model. In addition, if informed traders perfectly observe the asset value v, one gets the Holden and Subrahmanyam (1992) model with homogeneously informed traders, while the Kyle (1985) model is obtained for the special case where there is only one informed trader.

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IV. Numerical Analysis

The algorithm for solving the model minimizes the mean squared error of the difference between the desired period zero (co)variances and the period zero (co)variances that result from running the backward induction. Since the algorithm requires good starting values, several global optimizers are used to reach convergence.[11]

Numerically, correlations have to be bounded above in absolute value to ensure that the covariance matrix $\mathbf{\Sigma_0}$ in (1) is positive definite. Denoting $\rho_0^{\bar{x},\bar{y}}$ as the unconditional correlation coefficient between two arbitrary variables $\bar{x}$ and $\bar{y}$, the absolute values of $\rho_0^{v,s_{i,0}}$, ρ_0^{v,u_n}, $\rho_0^{s_{i,0},u_n}$, $\rho_0^{s_{i,0},s_{j,0}}$, and $\rho_0^{u_n,u_{n+1}}$ have to be bounded above for the model to be well-defined. This suggests that the equilibrium in Proposition 1 is only sustainable for small to moderate correlations.

This study is interested in how the correlation structure of uninformed orders affects the equilibrium outcome. Thus, the main variables of interest are ρ_0^{v,u_n}, the unconditional correlation between uninformed orders and the asset value, and $\rho_0^{s_{i,0},u_n}$, the unconditional correlation between uninformed orders and informed traders' signals. In what follows, this paper fixes the number of informed traders to $M = 3$ and the number of trading periods to $N = 10$. Initial variances equal $\Sigma_0^v = 5$, $\Sigma_0^{s_{i,0}} = 1$, and $\Sigma_0^{u_n} = 3/N$.

Moreover, the unconditional informativeness of signals equals $\rho_0^{v,s_{i,0}} = 0.4$ and the unconditional autocorrelation in the uninformed order flow is set to $\rho_0^{u_n,u_{n+1}} = 0.1$ since positive autocorrelation is empirically the most relevant case.[12] Finally, three different unconditional signal correlations are considered: positive correlation $\rho_0^{s_{i,0},s_{j,0}} = 0.25$, zero correlation $\rho_0^{s_{i,0},s_{j,0}} = 0$, and negative correlation $\rho_0^{s_{i,0},s_{j,0}} = -0.25$. For each unconditional signal correlation different combinations of ρ_0^{v,u_n} and $\rho_0^{s_{i,0},u_n}$ are analyzed.

[11] This paper uses the MATLAB global optimizers "patternsearch," "GlobalSearch," and "MultiStart." In all numerical results the mean squared error is less than 1e-9.

[12] See Hasbrouck (1991a,b) and Brogaard, Hendershott, and Riordan (2019).

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A. Results

A.1. Positive Initial Signal Correlation

In this analysis the initial signal correlation equals $\rho_0^{s_{i,0},s_{j,0}} = 0.25$. For uninformed orders the initial correlation structures are ($\rho_0^{v,u_n} = 0.25$, $\rho_0^{s_{i,0},u_{n,0}} = 0.25$), ($\rho_0^{v,u_n} = 0.1$, $\rho_0^{s_{i,0},u_n} = -0.1$), ($\rho_0^{v,u_n} = -0.1$, $\rho_0^{s_{i,0},u_n} = 0.1$), and ($\rho_0^{v,u_n} = -0.25$, $\rho_0^{s_{i,0},u_n} = -0.25$).

Figure 1 plots informed traders' trading intensity β_n, the market maker's liquidity parameter λ_n, the conditional asset variance Σ_n^v, and informed traders' terminal conditional expected profits over time. If uninformed orders are initially positively correlated with both the asset value and signals, the market maker's price sensitivity monotonically decreases over time since most information about the asset value gets incorporated into prices in the first periods. In this scenario uninformed orders reveal information about the asset's true value, which generates competition for informed traders. Consequently, informed traders make the lowest expected profits and decrease trading intensities throughout the trading day.

If uninformed orders are initially positively correlated with the asset value but negatively correlated with signals, informed traders' trading intensity monotonically increases over time, even though uninformed traders still reveal information about the true asset value. Surprisingly, at the end of the trading day price discovery is highest in this scenario. The reason for this is that aggregate order flows remain consistently informative about the asset value, resulting in most information getting incorporated into prices. Consequently, the market maker's price sensitivity only slowly decreases over time.

If uninformed orders are initially negatively correlated with the asset value but positively correlated with signals, uninformed traders push the price away from the true asset value, which provides camouflage for informed traders. As a result, informed traders trade most aggressively on their residual signals throughout the trading day and make high expected profits. Interestingly, the market maker's liquidity parameter exhibits a clear U-shape. It

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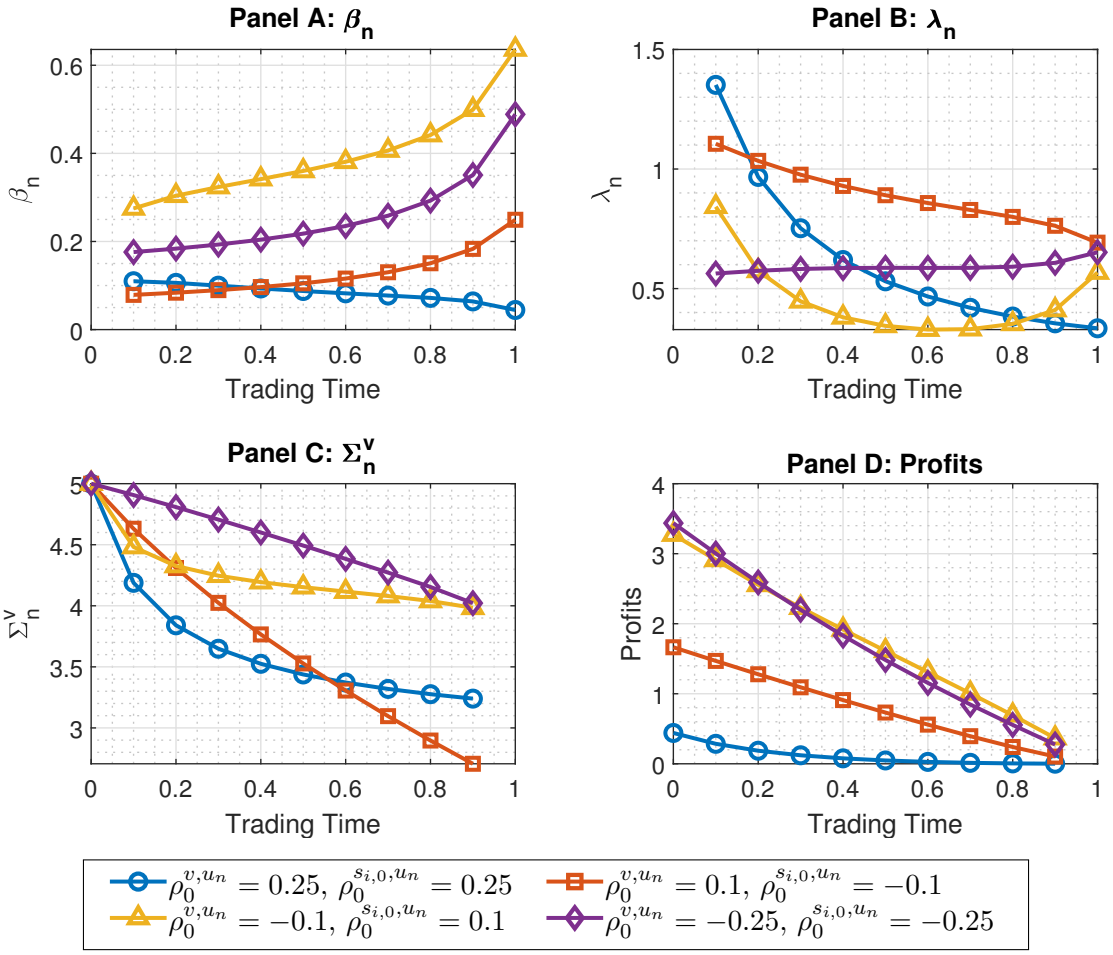


Figure 1: Trading Intensity, Liquidity Parameter, Conditional Asset Variance, and Expected Profits with Positive Initial Signal Correlation. The figure plots informed traders' trading intensity β_n, the market maker's liquidity parameter λ_n, the conditional asset variance Σ_n^v, and informed traders' terminal conditional expected profits over time. Parameter values are $M = 3$, $N = 10$, $\Sigma_0^v = 5$, $\Sigma_0^{s_{i,0}} = 1$, $\Sigma_0^{u_n} = 3/N$, $\rho_0^{v,s_{i,0}} = 0.4$, $\rho_0^{s_{i,0},s_{j,0}} = 0.25$, and $\rho_0^{u_n,u_{n+1}} = 0.1$. Model is solved for four initial parametrizations: ($\rho_0^{v,u_n} = 0.25$, $\rho_0^{s_{i,0},u_n} = 0.25$), ($\rho_0^{v,u_n} = 0.1$, $\rho_0^{s_{i,0},u_n} = -0.1$), ($\rho_0^{v,u_n} = -0.1$, $\rho_0^{s_{i,0},u_n} = 0.1$), and ($\rho_0^{v,u_n} = -0.25$, $\rho_0^{s_{i,0},u_n} = -0.25$).

initially declines because aggregate order flows are most informative in the first trading rounds when uninformed orders still contain information about signals. However, the liquidity parameter eventually increases in later periods because informed traders submit their most aggressive orders towards the end of the trading day.

Finally, if uninformed orders are initially negatively correlated with both the asset value and signals, informed traders also increase their trading intensity over time and make high expected profits. Similar to Kyle (1985), informed traders have an incentive to

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delay their trades towards later periods in order to capitalize on their private information. Moreover, the market maker can extract the least information from aggregate order flows since uninformed traders trade in the opposite direction of the true asset value and signals. Consequently, the market maker keeps a steady price sensitivity, resulting in slow price discovery over time.

Figure 2 shows the evolution of the conditional correlations $\rho_n^{u_{n+1},u_{n+2}}$, $\rho_n^{s_{i,0},s_{j,0}}$, $\rho_n^{v,u_{n+1}}$, and $\rho_n^{s_{i,0},u_{n+1}}$. If uninformed orders are initially positively correlated with both the asset value and signals, the conditional correlations monotonically decrease over time. Thus, uninformed orders become less informative about the asset value and signals in later periods. Furthermore, $\rho_n^{u_{n+1},u_{n+2}}$ strongly decreases in the first periods so that uninformed orders become conditionally less autocorrelated. Finally, $\rho_n^{s_{i,0},s_{j,0}}$ becomes negative over time, meaning that informed traders switch from trading in the same direction to trading in opposite directions. This is equivalent to informed traders developing a difference of opinion about the true asset value.

If uninformed orders are initially positively correlated with the asset value but negatively correlated with signals, the conditional signal correlation slowly decreases while the remaining conditional correlations stay approximately constant over time. Therefore, uninformed orders remain consistently informative about the asset value and exhibit a persistently high conditional autocorrelation. Moreover, the conditional signal correlation stays positive throughout the whole trading day so that no switch in trading directions occurs. Finally, note that $\rho_n^{v,u_{n+1}}$ is U-shaped and thus non-monotonic.

If uninformed orders are initially negatively correlated with the asset value but positively correlated with signals, the conditional correlations monotonically decline over time. In this scenario uninformed traders effectively conceal informed traders' orders from the market maker by pushing the price further away from the asset's true value while losing their informational value about signals after the first few trading rounds. Furthermore,

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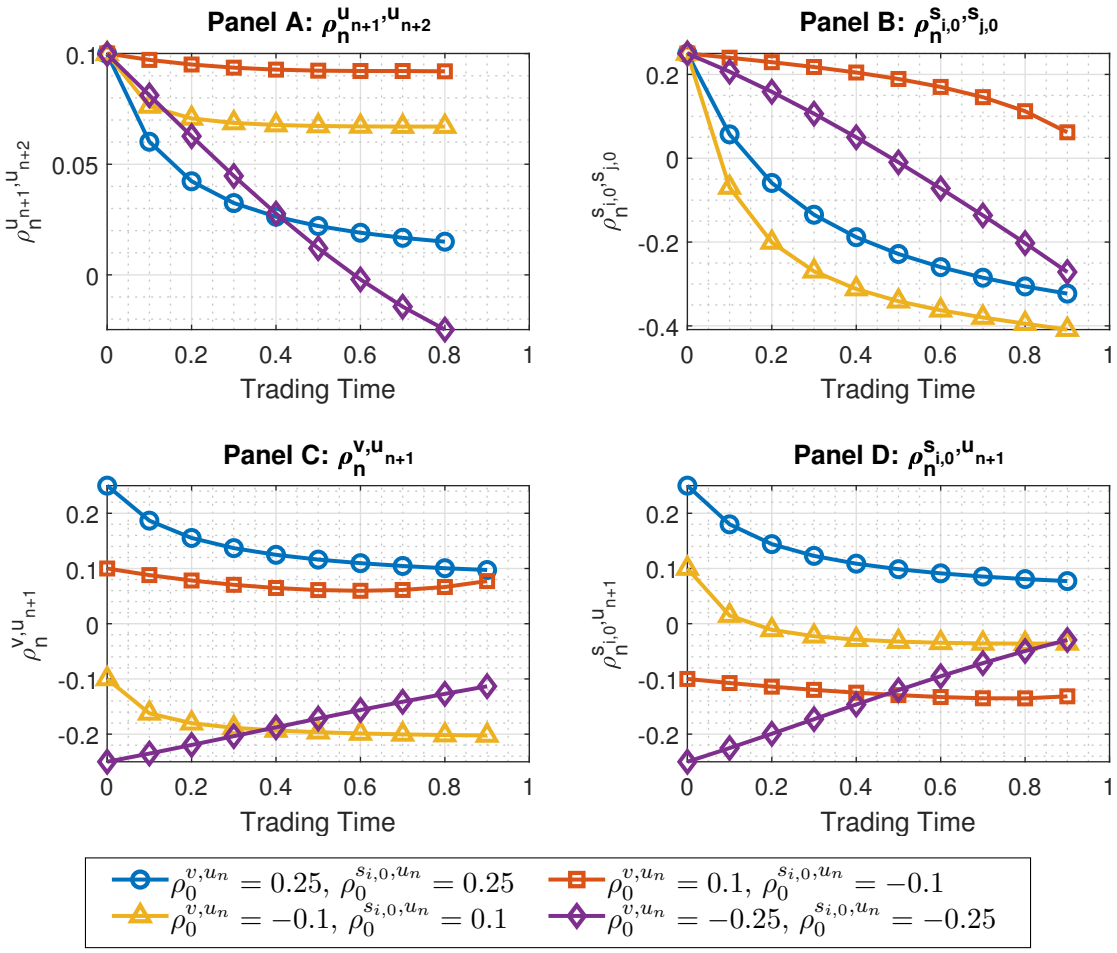


Figure 2: Conditional Correlations with Positive Initial Signal Correlation. The figure plots the conditional correlation between future uninformed orders $\rho_n^{u_{n+1},u_{n+2}}$, the conditional correlation between informed traders' signals $\rho_n^{s_{i,0},s_{j,0}}$, the conditional correlation between the asset value and future uninformed orders $\rho_n^{v,u_{n+1}}$, and the conditional correlation between informed traders' signals and future uninformed orders $\rho_n^{s_{i,0},u_{n+1}}$ over time. Parameter values are $M = 3$, $N = 10$, $\Sigma_0^v = 5$, $\Sigma_0^{s_{i,0}} = 1$, $\Sigma_0^{u_n} = 3/N$, $\rho_0^{v,s_{i,0}} = 0.4$, $\rho_0^{s_{i,0},s_{j,0}} = 0.25$, and $\rho_0^{u_n,u_{n+1}} = 0.1$. Model is solved for four initial parametrizations: ($\rho_0^{v,u_n} = 0.25$, $\rho_0^{s_{i,0},u_n} = 0.25$), ($\rho_0^{v,u_n} = 0.1$, $\rho_0^{s_{i,0},u_n} = -0.1$), ($\rho_0^{v,u_n} = -0.1$, $\rho_0^{s_{i,0},u_n} = 0.1$), and ($\rho_0^{v,u_n} = -0.25$, $\rho_0^{s_{i,0},u_n} = -0.25$).

the conditional signal correlation becomes negative after the first period. Thus, even if informed traders initially agree whether the asset is overvalued or undervalued, they will have different opinions after only one round of trading.

Finally, if uninformed orders are initially negatively correlated with both the asset value and signals, $\rho_n^{u_{n+1},u_{n+2}}$ and $\rho_n^{s_{i,0},s_{j,0}}$ monotonically decrease, while $\rho_n^{v,u_{n+1}}$ and $\rho_n^{s_{i,0},u_{n+1}}$ monotonically increase over time. Since uninformed orders become less negatively corre-

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lated with both the asset value and signals, uninformed traders provide less camouflage for informed traders in later periods. Surprisingly, the conditional uninformed order flow autocorrelation becomes negative over time. Thus, even if uninformed orders are positively autocorrelated, they can become negatively autocorrelated upon conditioning.

A.2. Zero Initial Signal Correlation

In this analysis the initial signal correlation equals $\rho_0^{s_{i,0},s_{j,0}} = 0$. For uninformed orders the initial correlation structures are ($\rho_0^{v,u_n} = 0.2$, $\rho_0^{s_{i,0},u_n} = 0.1$), ($\rho_0^{v,u_n} = 0.1$, $\rho_0^{s_{i,0},u_n} = -0.1$), ($\rho_0^{v,u_n} = -0.1$, $\rho_0^{s_{i,0},u_n} = 0.1$), and ($\rho_0^{v,u_n} = -0.1$, $\rho_0^{s_{i,0},u_n} = -0.2$).

Figure 3 plots informed traders' trading intensity β_n, the market maker's liquidity parameter λ_n, the conditional asset variance Σ_n^v, and informed traders' terminal conditional expected profits over time. It is worth noting that, with the exception of the case ($\rho_0^{v,u_n} = -0.1$, $\rho_0^{s_{i,0},u_n} = 0.1$), trading intensities are approximately unaffected by the correlation structure of uninformed orders if informed traders receive independent signals. In addition, trading intensities monotonically increase over time.

If uninformed orders are initially positively correlated with both the asset value and signals, informed traders make the lowest expected profits. Since uninformed traders reveal information about the true asset value, competition for informed traders is high. Moreover, the market maker's price sensitivity monotonically decreases since aggregate order flows become less informative about the asset value over time. Consequently, price discovery is highest in the first trading periods.

If uninformed orders are initially positively correlated with the asset value but negatively correlated with signals, informed traders make medium expected profits. In addition, the market maker's price sensitivity remains approximately constant throughout the trading day. In this scenario aggregate order flows stay consistently informative about the asset value, which is why price discovery is highest at the end of the trading day.

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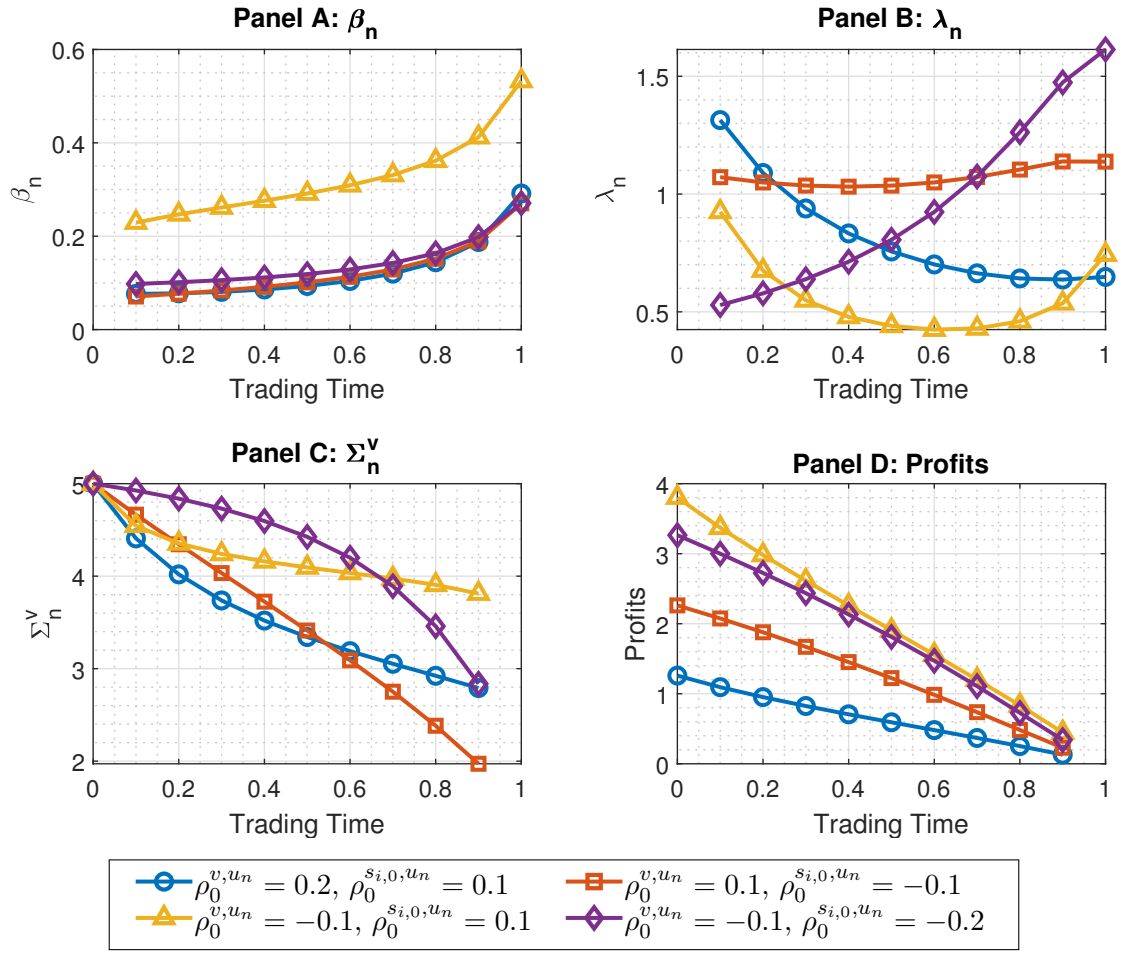


Figure 3: Trading Intensity, Liquidity Parameter, Conditional Asset Variance, and Expected Profits with Zero Initial Signal Correlation. The figure plots informed traders' trading intensity β_n, the market maker's liquidity parameter λ_n, the conditional asset variance Σ_n^v, and informed traders' terminal conditional expected profits over time. Parameter values are $M = 3$, $N = 10$, $\Sigma_0^v = 5$, $\Sigma_0^{s_{i,0}} = 1$, $\Sigma_0^{u_n} = 3/N$, $\rho_0^{v,s_{i,0}} = 0.4$, $\rho_0^{s_{i,0},s_{j,0}} = 0$, and $\rho_0^{u_n,u_{n+1}} = 0.1$. Model is solved for four initial parametrizations: ($\rho_0^{v,u_n} = 0.2$, $\rho_0^{s_{i,0},u_n} = 0.1$), ($\rho_0^{v,u_n} = 0.1$, $\rho_0^{s_{i,0},u_n} = -0.1$), ($\rho_0^{v,u_n} = -0.1$, $\rho_0^{s_{i,0},u_n} = 0.1$), and ($\rho_0^{v,u_n} = -0.1$, $\rho_0^{s_{i,0},u_n} = -0.2$).

If uninformed orders are initially negatively correlated with the asset value but positively correlated with signals, informed traders trade most aggressively on their residual signals and make high expected profits. The reason for this is that uninformed traders push the price away from the true asset value while not being informative about signals after the first few trading rounds. As a result, informed traders can effectively conceal their information from the market maker, which is why price discovery is lowest at the end of the trading day. Moreover, the liquidity parameter exhibits a clear U-shape. It

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initially declines because aggregate order flows contain most information about the asset value in the first trading rounds. However, the liquidity parameter eventually increases in later periods since informed traders trade most aggressively on their private information towards the end of the trading day.

Finally, if uninformed orders are initially negatively correlated with both the asset value and signals, informed traders also make high expected profits. Furthermore, the liquidity parameter monotonically increases over time. In this scenario uninformed orders provide camouflage for informed traders by trading in the opposite direction of the true asset value and signals. As a result, the market maker can extract the least information from aggregate order flows in the first periods, which is why she increases her price sensitivity throughout the trading day.

Figure 4 shows the evolution of the conditional correlations $\rho_n^{u_{n+1},u_{n+2}}$, $\rho_n^{s_{i,0},s_{j,0}}$, $\rho_n^{v,u_{n+1}}$, and $\rho_n^{s_{i,0},u_{n+1}}$. If uninformed orders are initially positively correlated with both the asset value and signals, the conditional correlations monotonically decrease over time. Therefore, uninformed orders become less informative about the asset value and signals in later periods. Moreover, informed traders develop a difference of opinion about the asset value after only one round of trading and the conditional uninformed order flow autocorrelation strongly declines over time.

If uninformed orders are initially positively correlated with the asset value but negatively correlated with signals, the conditional uninformed order flow autocorrelation slowly decreases while the remaining conditional correlations stay approximately constant over time. In this scenario aggregate order flows remain consistently informative about the asset value, resulting in most information getting incorporated into prices at the end of the trading day. Moreover, uninformed orders stay conditionally highly autocorrelated and informed traders submit independent orders throughout most of the trading day. Finally, note that the conditional correlation between the asset value and

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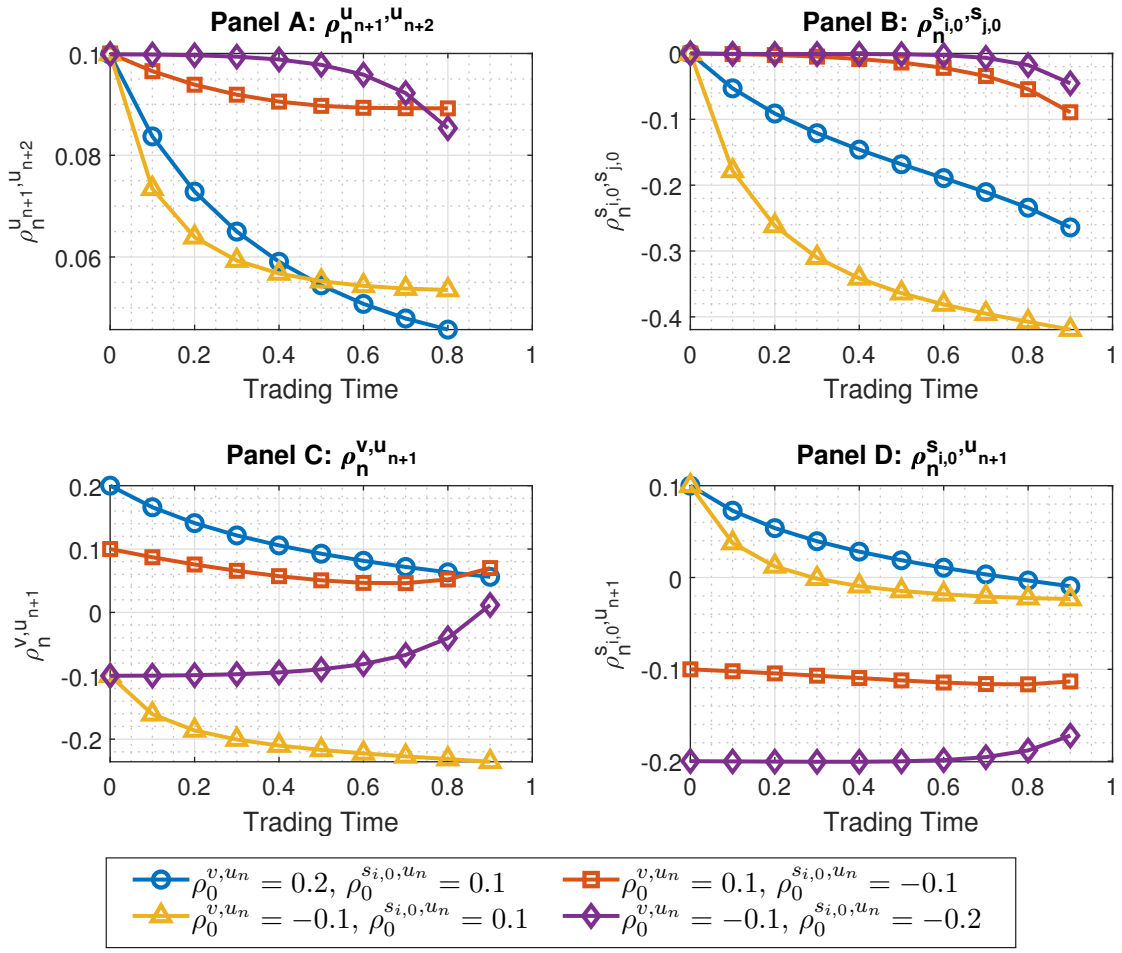


Figure 4: Conditional Correlations with Zero Initial Signal Correlation. The figure plots the conditional correlation between future uninformed orders $\rho_n^{u_{n+1},u_{n+2}}$, the conditional correlation between informed traders' signals $\rho_n^{s_{i,0},s_{j,0}}$, the conditional correlation between the asset value and future uninformed orders $\rho_n^{v,u_{n+1}}$, and the conditional correlation between informed traders' signals and future uninformed orders $\rho_n^{s_{i,0},u_{n+1}}$ over time. Parameter values are $M = 3$, $N = 10$, $\Sigma_0^v = 5$, $\Sigma_0^{s_{i,0}} = 1$, $\Sigma_0^{u_n} = 3/N$, $\rho_0^{v,s_{i,0}} = 0.4$, $\rho_0^{s_{i,0},s_{j,0}} = 0$, and $\rho_0^{u_n,u_{n+1}} = 0.1$. Model is solved for four initial parametrizations: ($\rho_0^{v,u_n} = 0.2$, $\rho_0^{s_{i,0},u_n} = 0.1$), ($\rho_0^{v,u_n} = 0.1$, $\rho_0^{s_{i,0},u_n} = -0.1$), ($\rho_0^{v,u_n} = -0.1$, $\rho_0^{s_{i,0},u_n} = 0.1$), and ($\rho_0^{v,u_n} = -0.1$, $\rho_0^{s_{i,0},u_n} = -0.2$).

uninformed orders is U-shaped and thus non-monotonic.

If uninformed orders are initially negatively correlated with the asset value but positively correlated with signals, the conditional correlations monotonically decrease over time. In this scenario uninformed orders provide high camouflage for informed traders by pushing the price further away from the true asset value while losing their informational value about signals after the first few trading rounds. Moreover, the conditional uninformed

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order flow autocorrelation strongly declines over time and informed traders develop a strong difference of opinion about the asset value.

Finally, if uninformed orders are initially negatively correlated with both the asset value and signals, the conditional correlations stay approximately constant over time and only change towards the end of the trading day. Since $\rho_n^{v,u_{n+1}}$ and $\rho_n^{s_{i,0},u_{n+1}}$ remain persistently negative, price discovery is low in the first trading rounds. In addition, uninformed orders stay conditionally highly autocorrelated and informed traders submit independent orders throughout most of the trading day.

A.3. Negative Initial Signal Correlation

In this analysis the initial signal correlation equals $\rho_0^{s_{i,0},s_{j,0}} = -0.25$. For uninformed orders the initial correlation structures are ($\rho_0^{v,u_n} = 0.4$, $\rho_0^{s_{i,0},u_n} = 0.15$), ($\rho_0^{v,u_n} = 0.2$, $\rho_0^{s_{i,0},u_n} = 0.1$), ($\rho_0^{v,u_n} = 0$, $\rho_0^{s_{i,0},u_n} = 0$), and ($\rho_0^{v,u_n} = -0.2$, $\rho_0^{s_{i,0},u_n} = -0.1$).[13]

Figure 5 plots informed traders' trading intensity β_n, the market maker's liquidity parameter λ_n, the conditional asset variance Σ_n^v, and informed traders' terminal conditional expected profits over time. If uninformed orders are initially positively correlated with both the asset value and signals, informed traders make the lowest expected profits and price discovery is highest. Since aggregate order flows are most informative about the asset value in the first trading periods, the market maker's price sensitivity declines over time and only increases towards the end of the trading day if the initial correlation between uninformed orders and the asset value is not too high. Interestingly, informed traders' trading intensity has the shape of a smirk if uninformed orders are initially highly correlated with the asset value. Therefore, trading intensities need not be monotonic.

If uninformed orders are initially uncorrelated with the asset value and signals, informed

[13]Previously, cases where ρ_0^{v,u_n} is positive while $\rho_0^{s_{i,0},u_n}$ is negative and vice versa were also studied. However, if signals are initially negatively correlated, the covariance matrix $\mathbf{\Sigma_0}$ in (1) is only positive definite with the specified parametrization if both ρ_0^{v,u_n} and $\rho_0^{s_{i,0},u_n}$ have the same sign.

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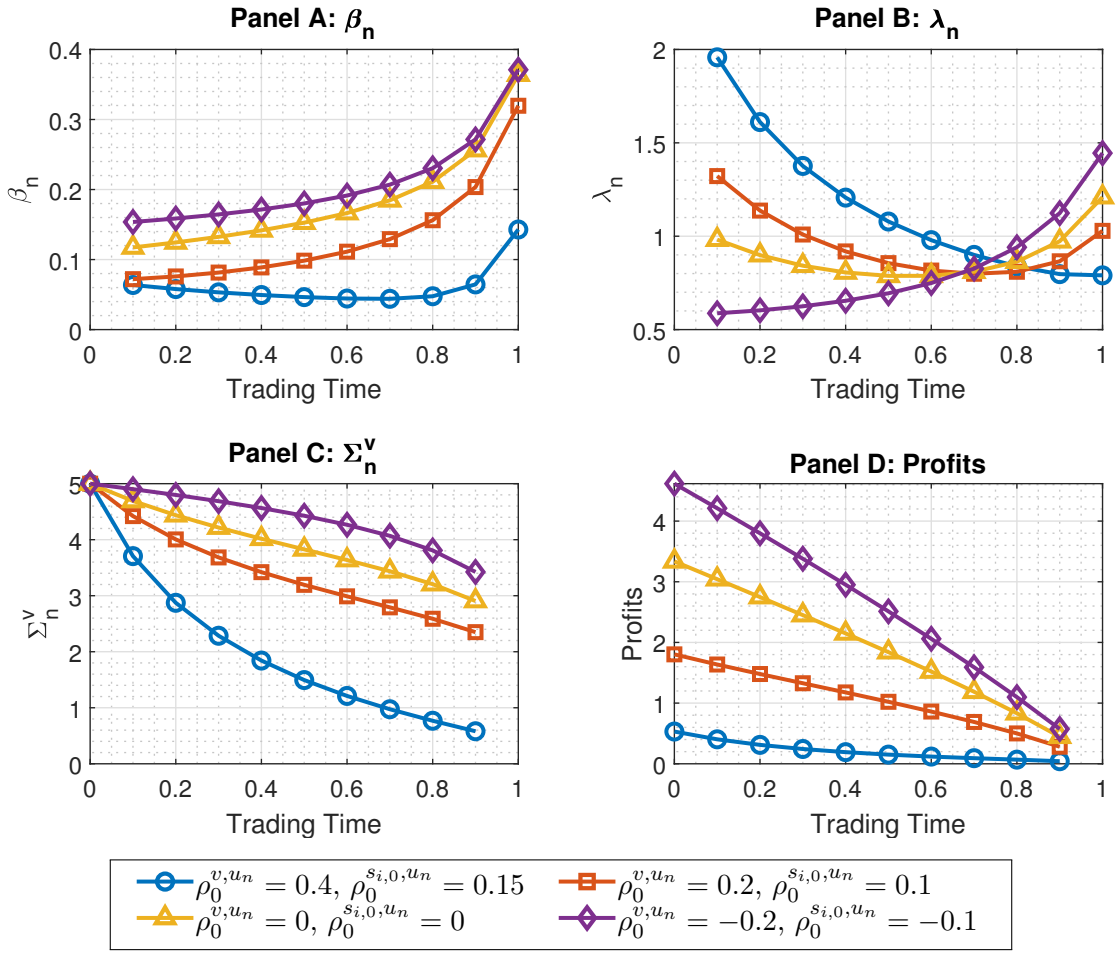


Figure 5: Trading Intensity, Liquidity Parameter, Conditional Asset Variance, and Expected Profits with Negative Initial Signal Correlation. The figure plots informed traders' trading intensity β_n, the market maker's liquidity parameter λ_n, the conditional asset variance Σ_n^v, and informed traders' terminal conditional expected profits over time. Parameter values are $M = 3$, $N = 10$, $\Sigma_0^v = 5$, $\Sigma_0^{s_{i,0}} = 1$, $\Sigma_0^{u_n} = 3/N$, $\rho_0^{v,s_{i,0}} = 0.4$, $\rho_0^{s_{i,0},s_{j,0}} = -0.25$, and $\rho_0^{u_n,u_{n+1}} = 0.1$. Model is solved for four initial parametrizations: $(\rho_0^{v,u_n} = 0.4, \rho_0^{s_{i,0},u_n} = 0.15)$, $(\rho_0^{v,u_n} = 0.2, \rho_0^{s_{i,0},u_n} = 0.1)$, $(\rho_0^{v,u_n} = 0, \rho_0^{s_{i,0},u_n} = 0)$, and $(\rho_0^{v,u_n} = -0.2, \rho_0^{s_{i,0},u_n} = -0.1)$.

traders trade aggressively on their private information and make high expected profits. In this scenario uninformed orders become conditionally negatively correlated with both the asset value and signals over time. Thus, uninformed traders effectively conceal informed traders' orders by pushing the price away from the true asset value. Furthermore, the market maker's price sensitivity exhibits a clear U-shape. It initially declines because aggregate order flows are most informative about the asset value in the first trading rounds. However, the liquidity parameter eventually increases in later periods because informed

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traders submit their most aggressive orders towards the end of the trading day.

Finally, if uninformed orders are initially negatively correlated with both the asset value and signals, informed traders trade most aggressively on their residual signals and make the highest expected profits. Moreover, the market maker's liquidity parameter monotonically increases over time and price discovery is lowest at the end of the trading day. In this scenario uninformed traders trade in the opposite direction of the true asset value and signals. As a result, informed traders can effectively conceal their orders from the market maker, who extracts the least information about the asset value from aggregate order flows.

Figure 6 shows the evolution of the conditional correlations $\rho_n^{u_{n+1},u_{n+2}}$, $\rho_n^{s_{i,0},s_{j,0}}$, $\rho_n^{v,u_{n+1}}$, and $\rho_n^{s_{i,0},u_{n+1}}$. If uninformed orders are initially positively correlated with both the asset value and signals, the conditional correlations, with the exception of $\rho_n^{v,u_{n+1}}$, decline over time. In particular, uninformed orders become conditionally less autocorrelated and informed traders' difference of opinion about the asset value gets reinforced throughout the trading day. In addition, uninformed orders become less informative about signals in later periods. Finally, $\rho_n^{v,u_{n+1}}$ has the shape of a smirk. In particular, uninformed orders can become even more informative about the asset value in later periods.

If uninformed orders are initially uncorrelated with both the asset value and signals, the conditional correlations monotonically decrease over time. Therefore, even if autocorrelated uninformed orders initially contain no information about the asset value and signals, they do so upon conditioning. In particular, uninformed orders become negatively correlated with the asset value and signals, providing camouflage for informed traders. Moreover, both the conditional uninformed order flow autocorrelation and the conditional signal correlation decline over time so that informed trader's initial difference of opinion about the asset value grows stronger throughout the trading day.

Finally, if uninformed orders are initially negatively correlated with both the asset value

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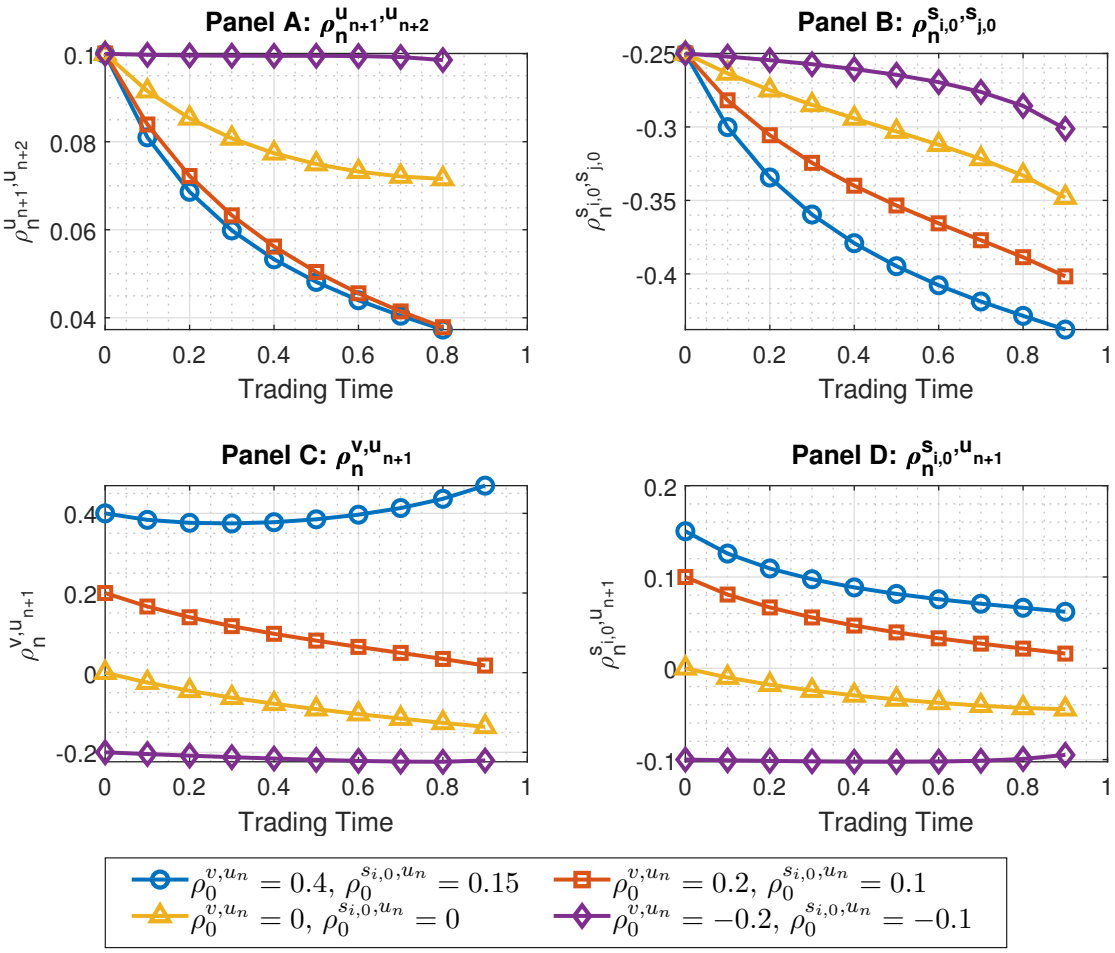


Figure 6: Conditional Correlations with Negative Initial Signal Correlation. The figure plots the conditional correlation between future uninformed orders $\rho_n^{u_{n+1},u_{n+2}}$, the conditional correlation between informed traders' signals $\rho_n^{s_{i,0},s_{j,0}}$, the conditional correlation between the asset value and future uninformed orders $\rho_n^{v,u_{n+1}}$, and the conditional correlation between informed traders' signals and future uninformed orders $\rho_n^{s_{i,0},u_{n+1}}$ over time. Parameter values are $M = 3$, $N = 10$, $\Sigma_0^v = 5$, $\Sigma_0^{s_{i,0}} = 1$, $\Sigma_0^{u_n} = 3/N$, $\rho_0^{v,s_{i,0}} = 0.4$, $\rho_0^{s_{i,0},s_{j,0}} = -0.25$, and $\rho_0^{u_n,u_{n+1}} = 0.1$. Model is solved for four initial parametrizations: $(\rho_0^{v,u_n} = 0.4,\ \rho_0^{s_{i,0},u_n} = 0.15)$, $(\rho_0^{v,u_n} = 0.2,\ \rho_0^{s_{i,0},u_n} = 0.1)$, $(\rho_0^{v,u_n} = 0,\ \rho_0^{s_{i,0},u_n} = 0)$, and $(\rho_0^{v,u_n} = -0.2,\ \rho_0^{s_{i,0},u_n} = -0.1)$.

and signals, the conditional signal correlation slowly decline while the remaining conditional correlations stay approximately constant over time. It is interesting to note that the conditional uninformed order flow autocorrelation stays persistently high. Moreover, uninformed orders remain persistently negatively correlated with both the asset value and signals. In this scenario informed traders can most effectively conceal their private information from the market maker and make the highest expected profits.

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V. Conclusion

This paper has derived a dynamic trading equilibrium with heterogeneously informed traders and (auto)correlated uninformed orders. An extensive numerical analysis has shown that (auto)correlated uninformed orders have significant effects on trading strategies, the very shape of the liquidity parameter, and information efficiency in markets. In particular, autocorrelation in the uninformed order flow implies uninformed orders to be conditionally correlated with the asset value and informed traders' signals in the cross-section.

Since informed traders have an intrinsic motive to submit unpredictable orders, the empirically observed autocorrelation in order flows is likely to come from uninformed traders. Given the non-trivial effects of (auto)correlated uninformed orders on the dynamic trading equilibrium, this paper wants to emphasize that strategic trading models should not assume uninformed traders to be simply noise.

The results in this paper provide testable implications for the dynamics of prices, order flows, and market depth. Moreover, this paper's model serves as a framework for interesting future research. Since this paper has studied a discrete-time framework, one natural extension is to solve the model in continuous time. Another extension is to drop the assumption of risk-neutral investors and allow for a more general utility function. Finally, the model only accounts for market orders and allowing traders to submit limit orders might provide new valuable insights.

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Appendix

A. Proof of Lemma 1

Proof. Assume informed orders are as in (5). Then for all $n \in \{1, \ldots, N\}$ one has

$$\begin{aligned} p_n &= \mathbb{E}\left[v \middle| y_1, \ldots, y_n\right] \\ &= p_{n-1} + \mathbb{E}\left[v - p_{n-1} \middle| y_1, \ldots, y_n\right] \\ &= p_{n-1} + \mathbb{E}\left[v - p_{n-1} \middle| y_1, \ldots, y_{n-1}, y_n - r_{n,n-1}\right] \\ &= p_{n-1} + \mathbb{E}\left[v - p_{n-1} \middle| y_n - r_{n,n-1}\right] \\ &= p_{n-1} + \lambda_n (y_n - r_{n,n-1}). \end{aligned} \tag{A1}$$

In deriving this expression, the second line follows by definition since p_{n-1} is in the market maker's information set after n rounds of trading. The third line uses the fact that the information sets $(y_1, \ldots, y_n)$ and $(y_1, \ldots, y_{n-1}, y_n - r_{n,n-1})$ are equivalent since $r_{n,n-1}$ is a function of $(y_1, \ldots, y_{n-1})$. Going from the third to the fourth line is a direct application of the projection theorem for Gaussian random variables, which implies that both $v - p_{n-1} = v - \mathbb{E}\left[v \middle| y_1, \ldots, y_{n-1}\right]$ and $y_n - r_{n,n-1} = y_n - \mathbb{E}\left[y_n \middle| y_1, \ldots, y_{n-1}\right]$ are independent of $(y_1, \ldots, y_{n-1})$. Finally, λ_n is the regression coefficient from the projection of $v - p_{n-1}$ on $y_n - r_{n,n-1}$.

Similarly, for all $i \in \{1, \ldots, M\}$ and all $n \in \{1, \ldots, N\}$ one has

$$\begin{aligned} t_{i,n} &= \mathbb{E}\left[s_{i,0} \middle| y_1, \ldots, y_n\right] \\ &= t_{i,n-1} + \mathbb{E}\left[s_{i,0} - t_{i,n-1} \middle| y_1, \ldots, y_{n-1}, y_n - r_{n,n-1}\right] \\ &= t_{i,n-1} + \mathbb{E}\left[s_{i,n-1} \middle| y_n - r_{n,n-1}\right] \\ &= t_{i,n-1} + \zeta_n (y_n - r_{n,n-1}), \end{aligned} \tag{A2}$$

where ζ_n is the regression coefficient from the projection of $s_{i,n-1}$ on $y_n - r_{n,n-1}$.

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Finally, for all $n \in \{1, \ldots, N-1\}$ and $t \in \{1, \ldots, N-n\}$ one has

$$\begin{aligned} r_{n+t,n} &= \mathbb{E}\left[u_{n+t} \middle| y_1, \ldots, y_n\right] \\ &= r_{n+t,n-1} + \mathbb{E}\left[u_{n+t} - r_{n+t,n-1} \middle| y_1, \ldots, y_{n-1}, y_n - r_{n,n-1}\right] \\ &= r_{n+t,n-1} + \mathbb{E}\left[u_{n+t} - r_{n+t,n-1} \middle| y_n - r_{n,n-1}\right] \\ &= r_{n,n-1} + \mathbb{E}\left[u_{n+1} - r_{n,n-1} \middle| y_n - r_{n,n-1}\right] \\ &= r_{n,n-1} + \theta_n (y_n - r_{n,n-1}), \end{aligned} \tag{A3}$$

where θ_n is the regression coefficient from the projection of $u_{n+1} - r_{n,n-1}$ on $y_n - r_{n,n-1}$. In going from the third to the fourth line, the distribution assumption (1) is used, which implies that, conditional on information up to period n, u_{n+t} and u_{n+1} have the same distribution for all $t \in \{1, \ldots, N-n\}$. Consequently, one also has $r_{n+t,n-1} = r_{n,n-1}$. □

B. Proof of Lemma 2

Proof. Define for all $n \in \{1, \ldots, N\}$ the following conditional (co)variances:

$$\Sigma_{n-1}^{y_n} \equiv \text{Var}\left[y_n \middle| y_1, \ldots, y_{n-1}\right], \tag{A4}$$

$$\Sigma_{n-1}^{s_{i,0}, y_n} \equiv \text{Cov}\left[s_{i,0}, y_n \middle| y_1, \ldots, y_{n-1}\right]. \tag{A5}$$

Moreover, let $\Sigma_{n-1}^{\boldsymbol{s}, y_n}$ be the $(M \times 1)$ vector with entries $\Sigma_{n-1}^{s_{i,0}, y_n}$. Then the property of multivariate Gaussian distributions implies that the update of the $(M \times M)$ covariance matrix $\Sigma_n^{\boldsymbol{s}}$ with diagonals $\Sigma_n^{s_{i,0}}$ and off-diagonals $\Sigma_n^{s_{i,0}, s_{j,0}}$ can be expressed recursively as

$$\Sigma_n^{\boldsymbol{s}} = \Sigma_{n-1}^{\boldsymbol{s}} - \Sigma_{n-1}^{\boldsymbol{s}, y_n} \left(\Sigma_{n-1}^{y_n}\right)^{-1} \left(\Sigma_{n-1}^{\boldsymbol{s}, y_n}\right)'. \tag{A6}$$

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Since the entries in $\Sigma_{n-1}^{\boldsymbol{s},y_n}$ are all identical, it can be concluded that

$$\Sigma_{n-1}^{s_{i,0}} - \Sigma_n^{s_{i,0}} = \Sigma_{n-1}^{s_{i,0},s_{j,0}} - \Sigma_n^{s_{i,0},s_{j,0}}. \tag{A7}$$

A similar analysis yields

$$\Sigma_{n-1}^{u_n} - \Sigma_n^{u_{n+1}} = \Sigma_{n-1}^{u_n,u_{n+1}} - \Sigma_n^{u_{n+1},u_{n+2}}. \tag{A8}$$

Finally, to show the relation for λ_n, note that according to (7):

$$\begin{aligned}
&\lambda_n(y_n - r_{n,n-1}) \\
=&p_n - p_{n-1} \\
=&\mathbb{E}\left[v - p_{n-1}\middle| y_1, \ldots, y_n\right] \\
=&\mathbb{E}\left[\mathbb{E}\left[v - p_{n-1}\middle| s_{1,0}, \ldots, s_{M,0}, u_n, y_1, \ldots, y_{n-1}\right]\middle| y_1, \ldots, y_n\right] \\
=&\mathbb{E}\left[\mathbb{E}\left[v - p_{n-1}\middle| s_{1,n-1}, \ldots, s_{M,n-1}, u_n - r_{n,n-1}, y_1, \ldots, y_{n-1}\right]\middle| y_1, \ldots, y_{n-1}, y_n - r_{n,n-1}\right] \\
=&\mathbb{E}\left[\mathbb{E}\left[v - p_{n-1}\middle| s_{1,n-1}, \ldots, s_{M,n-1}, u_n - r_{n,n-1}\right]\middle| y_n - r_{n,n-1}\right] \\
=&\mathbb{E}\left[\begin{pmatrix} \Sigma_{n-1}^{v,\boldsymbol{s}} & \Sigma_{n-1}^{v,u_n} \end{pmatrix} \begin{pmatrix} \Sigma_{n-1}^{\boldsymbol{s}} & \Sigma_{n-1}^{\boldsymbol{s},u_n} \\ \Sigma_{n-1}^{u_n,\boldsymbol{s}} & \Sigma_{n-1}^{u_n} \end{pmatrix}^{-1} \begin{pmatrix} s_{1,n-1} \\ \vdots \\ s_{M,n-1} \\ u_n - r_{n,n-1} \end{pmatrix} \middle| y_n - r_{n,n-1}\right] \\
=&\begin{pmatrix} \Sigma_{n-1}^{v,\boldsymbol{s}} & \Sigma_{n-1}^{v,u_n} \end{pmatrix} \begin{pmatrix} \Sigma_{n-1}^{\boldsymbol{s}} & \Sigma_{n-1}^{\boldsymbol{s},u_n} \\ \Sigma_{n-1}^{u_n,\boldsymbol{s}} & \Sigma_{n-1}^{u_n} \end{pmatrix}^{-1} \begin{pmatrix} \zeta_n \\ \vdots \\ \zeta_n \\ \tilde{\theta}_n \end{pmatrix} (y_n - r_{n,n-1}).
\end{aligned} \tag{A9}$$

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In deriving the above expression, the fourth line follows from the law of iterated expectations since the information set $(y_1, \ldots, y_n)$ is a subset of $(s_{1,0}, \ldots, s_{M,0}, u_n, y_1, \ldots, y_{n-1})$. Also, note that the information set $(s_{1,0}, \ldots, s_{M,0}, u_n, y_1, \ldots, y_{n-1})$ can be equivalently expressed as $(s_{1,n-1}, \ldots, s_{M,n-1}, u_n - r_{n,n-1}, y_1, \ldots, y_{n-1})$, while $(y_1, \ldots, y_n)$ is equivalent to $(y_1, \ldots, y_{n-1}, y_n - r_{n,n-1})$. Then the projection theorem for Gaussian random variables implies that $v - p_{n-1}$, $(s_{1,n-1}, \ldots, s_{M,n-1}, u_n - r_{n,n-1})$, and $y_n - r_{n,n-1}$ are all independent of $(y_1, \ldots, y_{n-1})$. Finally, the last two lines follow from the property of multivariate Gaussian distributions and $\tilde{\theta}_n$ is the regression coefficient from the projection of $u_n - r_{n,n-1}$ on $y_n - r_{n,n-1}$. □

C. Proof of Lemma 3

Proof. The proof is analogous to Foster and Viswanathan (1996). In every period $n-1$ where $n \in \{1, \ldots, N\}$ every informed trader $i \in \{1, \ldots, M\}$ forecasts the next period's uninformed order flow u_n as follows:

$$\begin{aligned}
\mathbb{E}\left[u_n \middle| s_{i,0}, y_1, \ldots, y_{n-1}, x_{i,1}, \ldots, x_{i,n-1}\right] &= \mathbb{E}\left[u_n \middle| s_{i,0}, y_1, \ldots, y_{n-1}\right] \\
&= r_{n,n-1} + \mathbb{E}\left[u_n - r_{n,n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}\right] \\
&= r_{n,n-1} + \mathbb{E}\left[u_n - r_{n,n-1} \middle| s_{i,n-1}, y_1, \ldots, y_{n-1}\right] \\
&= r_{n,n-1} + \mathbb{E}\left[u_n - r_{n,n-1} \middle| s_{i,n-1}\right] \\
&= r_{n,n-1} + \psi_n s_{i,n-1},
\end{aligned} \tag{A10}$$

The first line exploits the result that the trading history $(x_{i,1}, \ldots, x_{i,n-1})$ is redundant information, as was shown in (16). The second line follows by definition since $r_{n,n-1}$ is in informed trader i's information set after $n-1$ periods of trading. The third line uses the fact that the information sets $(s_{i,0}, y_1, \ldots, y_{n-1})$ and $(s_{i,n-1}, y_1, \ldots, y_{n-1})$ are equivalent since $s_{i,n-1} = s_{i,0} - t_{i,n-1}$ and $t_{i,n-1}$ is a function of $(y_1, \ldots, y_{n-1})$. Finally, the last two

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lines follow from the projection theorem for Gaussian random variables and ψ_n is the regression coefficient from the projection of $u_n - r_{n,n-1}$ on $s_{i,n-1}$.

Similarly, informed trader i's update of the asset value v is

$$\begin{aligned}\mathbb{E}\left[v\middle|s_{i,0}, y_1, \ldots, y_{n-1}, x_{i,1}, \ldots, x_{i,n-1}\right] &= p_{n-1} + \mathbb{E}\left[v - p_{n-1}\middle|s_{i,n-1}\right] \\ &= p_{n-1} + \eta_n s_{i,n-1},\end{aligned} \tag{A11}$$

where η_n is the regression coefficient from the projection of $v - p_{n-1}$ on $s_{i,n-1}$.

Moreover, informed trader i's belief about informed trader j's signal $s_{j,0}$, where $j \in \{1, \ldots, M\}$ and $j \neq i$, equals

$$\begin{aligned}\mathbb{E}\left[s_{j,0}\middle|s_{i,0}, y_1, \ldots, y_{n-1}, x_{i,1}, \ldots, x_{i,n-1}\right] &= t_{j,n-1} + \mathbb{E}\left[s_{j,n-1}\middle|s_{i,n-1}\right] \\ &= t_{j,n-1} + \phi_n s_{i,n-1},\end{aligned} \tag{A12}$$

where ϕ_n is the regression coefficient from the projection of $s_{j,n-1}$ on $s_{i,n-1}$. □

D. Proof of Lemma 4

Proof. The proof is taken from Foster and Viswanathan (1996) and is shown by induction. Consider the aggregate order flow in the first trading period as the base case:

$$y_1 = \sum_{j \neq i} \beta_1 s_{j,0} + \tilde{x}_{i,1} + u_1 = \sum_{j \neq i} \beta_1 \hat{s}^i_{j,0} + \beta_1 \hat{s}^i_{i,0} + (\tilde{x}_{i,1} - \beta_1 \hat{s}^i_{i,0}) + u_1 = \hat{y}^i_1 + (\tilde{x}_{i,1} - \beta_1 s_{i,0}). \tag{A13}$$

Since $(s_{i,0}, y_1, \tilde{x}_{i,1})$ is in informed trader i's information set after the first trading period, so is $\hat{y}^i_1$.

Assume as induction hypothesis that $(\hat{y}^i_1, \ldots, \hat{y}^i_{n-1})$ is in informed trader i's information set after $n-1$ periods where $n \in \{2, \ldots, N\}$. To show that $\hat{y}^i_n$ is in informed trader i's

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information set after n periods, write the aggregate order flow in period n as

$$\begin{aligned} y_n &= \sum_{j\neq i} \beta_n s_{j,n-1} + \tilde{x}_{i,n} + u_n \\ &= \sum_{j\neq i} \beta_n (s_{j,0} - t_{j,n-1}) + \tilde{x}_{i,n} + u_n \\ &= \sum_{j\neq i} \beta_n \left(s_{j,0} - \hat{t}^i_{j,n-1}\right) + \sum_{j\neq i} \beta_n \left(\hat{t}^i_{j,n-1} - t_{j,n-1}\right) + \tilde{x}_{i,n} + u_n \\ &= \sum_{j\neq i} \beta_n \hat{s}^i_{j,n-1} + \beta_n \hat{s}^i_{i,n-1} + u_n + \sum_{j\neq i} \beta_n \left(\hat{t}^i_{j,n-1} - t_{j,n-1}\right) + \tilde{x}_{i,n} - \beta_n \hat{s}^i_{i,n-1} \\ &= \hat{y}^i_n + \sum_{j\neq i} \beta_n \left(\hat{t}^i_{j,n-1} - t_{j,n-1}\right) + \tilde{x}_{i,n} - \beta_n \left(s_{i,0} - \hat{t}^i_{i,n-1}\right). \end{aligned} \tag{A14}$$

Note that $\hat{t}^i_{j,n-1}$ is a function of the equilibrium aggregate order flow history $(\hat{y}^i_1, \ldots, \hat{y}^i_{n-1})$, which by induction hypothesis is in informed trader i's information set. Since informed trader i knows $(s_{i,0}, y_1, \ldots, y_n, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n})$ and thus also $t_{j,n-1}$ after n trading periods, it can be concluded that $\hat{y}^i_n$ is also in informed trader i's information set. □

E. Proof of Lemma 5

Proof. The proof is analogous to Foster and Viswanathan (1996). Suppose informed trader i has submitted an arbitrary order sequence $(\tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1})$ in the first $n-1$ periods where $n \in \{1, \ldots, N\}$. Then

$$\begin{aligned} &\mathbb{E}\left[u_n - r_{n,n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] \\ =&\mathbb{E}\left[u_n - \hat{r}^i_{n,n-1} + \hat{r}^i_{n,n-1} - r_{n,n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] \\ =&\mathbb{E}\left[u_n - \hat{r}^i_{n,n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] + \left(\hat{r}^i_{n,n-1} - r_{n,n-1}\right). \end{aligned} \tag{A15}$$

Note that Lemma 4 has shown that $(\hat{y}^i_1, \ldots, \hat{y}^i_{n-1})$ can be recursively constructed from the information set $(s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1})$, which implies that $\hat{r}^i_{n,n-1} - r_{n,n-1}$ is in informed trader i's information set after $n-1$ periods. Moreover, the term on the

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left-hand side equals

$$\begin{aligned}
&\mathbb{E}\left[u_n - \hat{r}^i_{n,n-1}\middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] \\
=&\mathbb{E}\left[u_n - \hat{r}^i_{n,n-1}\middle| s_{i,0}, \hat{y}^i_1, \ldots, \hat{y}^i_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] \\
=&\mathbb{E}\left[u_n - \hat{r}^i_{n,n-1}\middle| \hat{s}^i_{i,n-1}, \hat{y}^i_1, \ldots, \hat{y}^i_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] \\
=&\mathbb{E}\left[u_n - \hat{r}^i_{n,n-1}\middle| \hat{s}^i_{i,n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] \\
=&\mathbb{E}\left[u_n - \hat{r}^i_{n,n-1}\middle| \hat{s}^i_{i,n-1}\right] \\
=&\psi_n \hat{s}^i_{i,n-1},
\end{aligned} \tag{A16}$$

where the penultimate line makes use of the fact that $(\tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1})$ is an arbitrary sequence of numbers that is unrelated to equilibrium play. Thus, it can be concluded that

$$\mathbb{E}\left[u_n - r_{n,n-1}\middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] = \psi_n \hat{s}^i_{i,n-1} + \left(\hat{r}^i_{n,n-1} - r_{n,n-1}\right). \tag{A17}$$

A similar analysis shows that

$$\mathbb{E}\left[v - p_{n-1}\middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] = \eta_n \hat{s}^i_{i,n-1} + \left(\hat{p}^i_{n-1} - p_{n-1}\right) \tag{A18}$$

and

$$\mathbb{E}\left[s_{j,n-1}\middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] = \phi_n \hat{s}^i_{i,n-1} + \left(\hat{t}^i_{j,n-1} - t_{j,n-1}\right), \tag{A19}$$

as was to be shown. □

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F. Proof of Proposition 1

Proof. Given the conjectured value function (25), informed trader i's Bellman equation in trading period $n \in \{1, \ldots, N\}$ writes

$$\begin{aligned}
&\max_{x_{i,n}} \mathbb{E}\left[(v - p_n)x_{i,n} \middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] \\
&+ \mathbb{E}\Big[a_n \left(\hat{s}^i_{i,n}\right)^2 + b_n \left(\hat{p}^i_n - p_n\right)^2 + c_n \left(\hat{t}^i_{j,n} - t_{j,n}\right)^2 + d_n \left(\hat{r}^i_{n+1,n} - r_{n+1,n}\right)^2 \\
&\quad + e_n \hat{s}^i_{i,n}\left(\hat{p}^i_n - p_n\right) + f_n \hat{s}^i_{i,n}\left(\hat{t}^i_{j,n} - t_{j,n}\right) + g_n \hat{s}^i_{i,n}\left(\hat{r}^i_{n+1,n} - r_{n+1,n}\right) + h_n \left(\hat{p}^i_n - p_n\right)\left(\hat{t}^i_{j,n} - t_{j,n}\right) \\
&\quad + i_n \left(\hat{p}^i_n - p_n\right)\left(\hat{r}^i_{n+1,n} - r_{n+1,n}\right) + j_n \left(\hat{t}^i_{j,n} - t_{j,n}\right)\left(\hat{r}^i_{n+1,n} - r_{n+1,n}\right) + k_n \\
&\quad \Big| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\Big].
\end{aligned} \tag{A20}$$

Under the conjectured equilibrium one has[14]

$$p_n = p_{n-1} + \lambda_n(y_n - r_{n,n-1}) = p_{n-1} + \lambda_n \left(x_{i,n} + \sum_{j \neq i} \beta_n s_{j,n-1} + u_n - r_{n,n-1}\right), \tag{A21}$$

$$t_{j,n} = t_{j,n-1} + \zeta_n(y_n - r_{n,n-1}) = t_{j,n-1} + \zeta_n \left(x_{i,n} + \sum_{j \neq i} \beta_n s_{j,n-1} + u_n - r_{n,n-1}\right), \tag{A22}$$

$$r_{n+1,n} = r_{n,n-1} + \theta_n(y_n - r_{n,n-1}) = r_{n,n-1} + \theta_n \left(x_{i,n} + \sum_{j \neq i} \beta_n s_{j,n-1} + u_n - r_{n,n-1}\right), \tag{A23}$$

[14] By definition, $r_{N+1,N} = 0$ since there are no uninformed orders after the final trading period.

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so that the first order condition with respect to $x_{i,n}$ yields

$$\begin{aligned}
&\mathbb{E}\left[v - p_{n-1} - \lambda_n\left(\sum_{j\neq i}\beta_n s_{j,n-1} + u_n - r_{n,n-1}\right)\middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] - 2\lambda_n x_{i,n} \\
&+\mathbb{E}\Big[2b_n\left(\hat{p}^i_n - p_n\right)(-\lambda_n) + 2c_n\left(\hat{t}^i_{j,n} - t_{j,n}\right)(-\zeta_n) + 2d_n\left(\hat{r}^i_{n+1,n} - r_{n+1,n}\right)(-\theta_n) \\
&\quad + e_n\hat{s}^i_{i,n}(-\lambda_n) + f_n\hat{s}^i_{i,n}(-\zeta_n) + g_n\hat{s}^i_{i,n}(-\theta_n) + h_n\left[-\lambda_n\left(\hat{t}^i_{j,n} - t_{j,n}\right) - \zeta_n\left(\hat{p}^i_n - p_n\right)\right] \\
&\quad + i_n\left[-\lambda_n\left(\hat{r}^i_{n+1,n} - r_{n+1,n}\right) - \theta_n\left(\hat{p}^i_n - p_n\right)\right] + j_n\left[-\zeta_n\left(\hat{r}^i_{n+1,n} - r_{n+1,n}\right) - \theta_n\left(\hat{t}^i_{j,n} - t_{j,n}\right)\right] \\
&\quad \Big| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\Big] = 0
\end{aligned} \tag{A24}$$

and the second order condition is

$$-\lambda_n(2-2b_n\lambda_n-h_n\zeta_n-i_n\theta_n)+\zeta_n(2c_n\zeta_n+h_n\lambda_n+j_n\theta_n)+\theta_n(2d_n\theta_n+i_n\lambda_n+j_n\zeta_n) < 0. \tag{A25}$$

Following Foster and Viswanathan (1996), expand the first order condition to obtain

$$\begin{aligned}
&\mathbb{E}\left[v - \hat{p}^i_{n-1} - \lambda_n\left(\sum_{j\neq i}\beta_n\hat{s}^i_{j,n-1} + u_n - \hat{r}^i_{n,n-1}\right)\middle| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right] \\
&\quad - 2\lambda_n x_{i,n} + \left(\hat{p}^i_{n-1} - p_{n-1}\right) + \lambda_n\left[\sum_{j\neq i}\beta_n\left(\hat{s}^i_{j,n-1} - s_{j,n-1}\right) - \left(\hat{r}^i_{n,n-1} - r_{n,n-1}\right)\right] \\
&-\mathbb{E}\Big[(e_n\lambda_n + f_n\zeta_n + g_n\theta_n)\hat{s}^i_{i,n} + (2b_n\lambda_n + h_n\zeta_n + i_n\theta_n)\left(\hat{p}^i_n - p_n\right) \\
&\quad + (2c_n\zeta_n + h_n\lambda_n + j_n\theta_n)\left(\hat{t}^i_{j,n} - t_{j,n}\right) + (2d_n\theta_n + i_n\lambda_n + j_n\zeta_n)\left(\hat{r}^i_{n+1,n} - r_{n+1,n}\right) \\
&\quad \Big| s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\Big] = 0,
\end{aligned} \tag{A26}$$

where $\hat{p}^i_{n-1} - p_{n-1}$ and $\hat{r}^i_{n,n-1} - r_{n,n-1}$ are in informed trader i's information set. Also,

$$\hat{s}^i_{j,n-1} - s_{j,n-1} = -\left(\hat{t}^i_{j,n-1} - t_{j,n-1}\right) \tag{A27}$$

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is in informed trader i's information set. Before proceeding with the first order condition, note that

$$\mathbb{E}\left[v-\hat{p}^i_{n-1}\middle|s_{i,0},y_1,\ldots,y_{n-1},\tilde{x}_{i,1},\ldots,\tilde{x}_{i,n-1}\right]=\mathbb{E}\left[v-\hat{p}^i_{n-1}\middle|\hat{s}^i_{i,n-1}\right]=\eta_n\hat{s}^i_{i,n-1}, \tag{A28}$$

$$\mathbb{E}\left[\hat{s}^i_{j,n-1}\middle|s_{i,0},y_1,\ldots,y_{n-1},\tilde{x}_{i,1},\ldots,\tilde{x}_{i,n-1}\right]=\mathbb{E}\left[\hat{s}^i_{j,n-1}\middle|\hat{s}^i_{i,n-1}\right]=\phi_n\hat{s}^i_{i,n-1}, \tag{A29}$$

$$\mathbb{E}\left[u_n-\hat{r}^i_{n,n-1}\middle|s_{i,0},y_1,\ldots,y_{n-1},\tilde{x}_{i,1},\ldots,\tilde{x}_{i,n-1}\right]=\mathbb{E}\left[u_n-\hat{r}^i_{n,n-1}\middle|\hat{s}^i_{i,n-1}\right]=\psi_n\hat{s}^i_{i,n-1}, \tag{A30}$$

where η_n, ϕ_n, and ψ_n are the corresponding regression coefficients. With these results one can show that

$$\begin{aligned}&\mathbb{E}\left[\hat{s}^i_{i,n}\middle|s_{i,0},y_1,\ldots,y_{n-1},\tilde{x}_{i,1},\ldots,\tilde{x}_{i,n-1}\right]\\ =&\hat{s}^i_{i,n-1}-\zeta_n\,\mathbb{E}\left[\beta_n\hat{s}^i_{i,n-1}+\sum_{j\neq i}\beta_n\hat{s}^i_{j,n-1}+u_n-\hat{r}^i_{n,n-1}\middle|\hat{s}^i_{i,n-1}\right]\\ =&\Big(1-\zeta_n\Big[\beta_n+(M-1)\beta_n\phi_n+\psi_n\Big]\Big)\hat{s}^i_{i,n-1}.\end{aligned} \tag{A31}$$

Moreover,

$$\begin{aligned}&\hat{p}^i_n-p_n=\hat{p}^i_{n-1}-p_{n-1}+\lambda_n\left[\hat{y}^i_n-\hat{r}^i_{n,n-1}-(y_n-r_{n,n-1})\right]\\ =&\hat{p}^i_{n-1}-p_{n-1}+\lambda_n\Big[\beta_n\hat{s}^i_{i,n-1}-x_{i,n}-(M-1)\beta_n\left(\hat{t}^i_{j,n-1}-t_{j,n-1}\right)-\left(\hat{r}^i_{n,n-1}-r_{n,n-1}\right)\Big].\end{aligned} \tag{A32}$$

Similarly,

$$\begin{aligned}&\hat{t}^i_{j,n}-t_{j,n}=\hat{t}^i_{j,n-1}-t_{j,n-1}+\zeta_n\left[\hat{y}^i_n-\hat{r}^i_{n,n-1}-(y_n-r_{n,n-1})\right]\\ =&\hat{t}^i_{j,n-1}-t_{j,n-1}+\zeta_n\Big[\beta_n\hat{s}^i_{i,n-1}-x_{i,n}-(M-1)\beta_n\left(\hat{t}^i_{j,n-1}-t_{j,n-1}\right)-\left(\hat{r}^i_{n,n-1}-r_{n,n-1}\right)\Big]\end{aligned} \tag{A33}$$

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and

$$\begin{aligned}&\hat{r}^i_{n+1,n} - r_{n+1,n} = \hat{r}^i_{n,n-1} - r_{n,n-1} + \theta_n\left[\hat{y}^i_n - \hat{r}^i_{n,n-1} - (y_n - r_{n,n-1})\right]\\ =&\hat{r}^i_{n,n-1} - r_{n,n-1} + \theta_n\Big[\beta_n\hat{s}^i_{i,n-1} - x_{i,n} - (M-1)\beta_n\left(\hat{t}^i_{j,n-1} - t_{j,n-1}\right) - \left(\hat{r}^i_{n,n-1} - r_{n,n-1}\right)\Big].\end{aligned} \tag{A34}$$

Using all of these relations in the first order condition, one can show that informed trader i's optimal strategy in period n is

$$x_{i,n} = \beta_n\hat{s}^i_{i,n-1} + \gamma_n\left(\hat{p}^i_{n-1} - p_{n-1}\right) + \alpha_n\left(\hat{t}^i_{j,n-1} - t_{j,n-1}\right) + \delta_n\left(\hat{r}^i_{n,n-1} - r_{n,n-1}\right), \tag{A35}$$

where the coefficients β_n, γ_n, α_n, and δ_n are just as in Proposition 1. Finally, plugging $x_{i,n}$ into the Bellman equation and noting that

$$\begin{aligned}&\mathbb{E}\left[\left(\hat{s}^i_{i,n}\right)^2\middle|s_{i,0}, y_1, \ldots, y_{n-1}, \tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n-1}\right]\\ =&\Big(\left[1-\zeta_n\beta_n\right]^2 - 2\zeta_n(1-\zeta_n\beta_n)\left[\psi_n + (M-1)\beta_n\phi_n\right]\Big)\left(\hat{s}^i_{i,n-1}\right)^2\\ &+\zeta_n^2\Big[\operatorname{Var}\left[u_n - \hat{r}^i_{n,n-1}\middle|s_{i,0}, \hat{y}^i_1, \ldots, \hat{y}^i_{n-1}\right] + 2(M-1)\beta_n\operatorname{Cov}\left[\hat{s}^i_{j,n-1}, u_n - \hat{r}^i_{n,n-1}\middle|s_{i,0}, \hat{y}^i_1, \ldots, \hat{y}^i_{n-1}\right]\\ &+(M-1)\beta_n^2\Big(\operatorname{Var}\left[\hat{s}^i_{j,n-1}\middle|s_{i,0}, \hat{y}^i_1, \ldots, \hat{y}^i_{n-1}\right] + (M-2)\operatorname{Cov}\left[\hat{s}^i_{j,n-1}, \hat{s}^i_{k,n-1}\middle|s_{i,0}, \hat{y}^i_1, \ldots, \hat{y}^i_{n-1}\right]\Big)\Big],\end{aligned} \tag{A36}$$

one obtains the value function coefficients for period $n-1$. At this stage of the proof the "hat" notation is suppressed because informed trader i will not deviate from his optimal strategy along the equilibrium path.

Note that η_n, ϕ_n, and ψ_n are the regression coefficients that result from the projection of $v - p_{n-1}$, $s_{j,n-1}$, and $u_n - r_{n,n-1}$ on $s_{i,n-1}$, respectively. Consequently,

$$\eta_n = \frac{\operatorname{Cov}\left[v - p_{n-1}, s_{i,n-1}\right]}{\operatorname{Var}\left[s_{i,n-1}\right]} = \frac{\Sigma^{v,s_{i,0}}_{n-1}}{\Sigma^{s_{i,0}}_{n-1}}, \tag{A37}$$

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$$\phi_n = \frac{\text{Cov}\left[s_{j,n-1}, s_{i,n-1}\right]}{\text{Var}\left[s_{i,n-1}\right]} = \frac{\Sigma_{n-1}^{s_{j,0},s_{i,0}}}{\Sigma_{n-1}^{s_{i,0}}}, \tag{A38}$$

$$\psi_n = \frac{\text{Cov}\left[u_n - r_{n,n-1}, s_{i,n-1}\right]}{\text{Var}\left[s_{i,n-1}\right]} = \frac{\Sigma_{n-1}^{u_n,s_{i,0}}}{\Sigma_{n-1}^{s_{i,0}}}. \tag{A39}$$

Next, one obtains

$$\begin{aligned}
&\text{Var}\left[u_n - r_{n,n-1}\middle| s_{i,0}, y_1, \ldots, y_{n-1}\right] \\
=&\mathbb{E}\left[(u_n - r_{n,n-1} - \psi_n s_{i,n-1})^2\right] \\
=&\mathbb{E}\left[(u_n - r_{n,n-1})^2\right] - 2\psi_n \mathbb{E}\left[(u_n - r_{n,n-1}) s_{i,n-1}\right] + \psi_n^2 \mathbb{E}\left[(s_{i,n-1})^2\right] \\
=&\Sigma_{n-1}^{u_n} - 2\psi_n \Sigma_{n-1}^{u_n,s_{i,0}} + \psi_n^2 \Sigma_{n-1}^{s_{i,0}} \\
=&\Sigma_{n-1}^{u_n} - \psi_n^2 \Sigma_{n-1}^{s_{i,0}}.
\end{aligned} \tag{A40}$$

Similarly,

$$\begin{aligned}
&\text{Var}\left[s_{j,n-1}\middle| s_{i,0}, y_1, \ldots, y_{n-1}\right] \\
=&\mathbb{E}\left[(s_{j,n-1} - \phi_n s_{i,n-1})^2\right] \\
=&\mathbb{E}\left[(s_{j,n-1})^2\right] - 2\phi_n \mathbb{E}\left[s_{j,n-1} s_{i,n-1}\right] + \phi_n^2 \mathbb{E}\left[(s_{i,n-1})^2\right] \\
=&\Sigma_{n-1}^{s_{j,0}} - 2\phi_n \Sigma_{n-1}^{s_{j,0},s_{i,0}} + \phi_n^2 \Sigma_{n-1}^{s_{i,0}} \\
=&(1 - \phi_n^2)\Sigma_{n-1}^{s_{i,0}}
\end{aligned} \tag{A41}$$

and

$$\begin{aligned}
&\text{Cov}\left[s_{j,n-1}, s_{k,n-1}\middle| s_{i,0}, y_1, \ldots, y_{n-1}\right] \\
=&\text{Cov}\left[s_{j,n-1} - \phi_n s_{i,n-1}, s_{k,n-1} - \phi_n s_{i,n-1}\right] \\
=&\text{Cov}\left[s_{j,n-1}, s_{i,n-1}\right] - \phi_n \text{Cov}\left[s_{j,n-1}, s_{i,n-1}\right] - \phi_n \text{Cov}\left[s_{k,n-1}, s_{i,n-1}\right] + \phi_n^2 \text{Cov}\left[s_{i,n-1}, s_{i,n-1}\right] \\
=&\Sigma_{n-1}^{s_{j,0},s_{i,0}} - 2\phi_n \Sigma_{n-1}^{s_{j,0},s_{i,0}} + \phi_n^2 \Sigma_{n-1}^{s_{i,0}} \\
=&(1 - \phi_n)\Sigma_{n-1}^{s_{j,0},s_{i,0}}
\end{aligned} \tag{A42}$$

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and

$$\begin{aligned}
&\operatorname{Cov}\left[s_{j,n-1}, u_n - r_{n,n-1} \middle| s_{i,0}, y_1, \ldots, y_{n-1}\right] \\
=&\operatorname{Cov}\left[s_{j,n-1} - \phi_n s_{i,n-1}, u_n - r_{n,n-1} - \psi_n s_{i,n-1}\right] \\
=&\operatorname{Cov}\left[s_{j,n-1}, u_n - r_{n,n-1}\right] - \phi_n \operatorname{Cov}\left[s_{i,n-1}, u_n - r_{n,n-1}\right] \\
&- \psi_n \operatorname{Cov}\left[s_{j,n-1}, s_{i,n-1}\right] + \phi_n\psi_n \operatorname{Cov}\left[s_{i,n-1}, s_{i,n-1}\right] \\
=&\Sigma_{n-1}^{s_{i,0},u_n} - \phi_n\Sigma_{n-1}^{s_{i,0},u_n} - \psi_n\Sigma_{n-1}^{s_{j,0},s_{i,0}} + \phi_n\psi_n\Sigma_{n-1}^{s_{i,0}} \\
=&(1-\phi_n)\Sigma_{n-1}^{s_{i,0},u_n}.
\end{aligned} \tag{A43}$$

To compute λ_n, recall that

$$y_n - r_{n,n-1} = \sum_{i=1}^{M} \beta_n s_{i,n-1} + u_n - r_{n,n-1}. \tag{A44}$$

Then the projection theorem implies

$$\begin{aligned}
\lambda_n &= \frac{\operatorname{Cov}\left[v - p_{n-1}, y_n - r_{n,n-1}\right]}{\operatorname{Var}\left[y_n - r_{n,n-1}\right]} \\
&= \frac{M\beta_n\Sigma_{n-1}^{v,s_{i,0}} + \Sigma_{n-1}^{v,u_n}}{M\beta_n^2\left[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\right] + \Sigma_{n-1}^{u_n} + 2M\beta_n\Sigma_{n-1}^{s_{i,0},u_n}}.
\end{aligned} \tag{A45}$$

Similarly,

$$\begin{aligned}
\zeta_n &= \frac{\operatorname{Cov}\left[s_{i,n-1}, y_n - r_{n,n-1}\right]}{\operatorname{Var}\left[y_n - r_{n,n-1}\right]} \\
&= \frac{\beta_n\left[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\right] + \Sigma_{n-1}^{s_{i,0},u_n}}{M\beta_n^2\left[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\right] + \Sigma_{n-1}^{u_n} + 2M\beta_n\Sigma_{n-1}^{s_{i,0},u_n}}
\end{aligned} \tag{A46}$$

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and[15]

$$\begin{aligned}\theta_n &= \frac{\text{Cov}\left[u_{n+1} - r_{n,n-1}, y_n - r_{n,n-1}\right]}{\text{Var}\left[y_n - r_{n,n-1}\right]} \\ &= \frac{M\beta_n \Sigma_{n-1}^{s_{i,0},u_n} + \Sigma_{n-1}^{u_n,u_{n+1}}}{M\beta_n^2\left[\Sigma_{n-1}^{s_{i,0}} + (M-1)\Sigma_{n-1}^{s_{i,0},s_{j,0}}\right] + \Sigma_{n-1}^{u_n} + 2M\beta_n \Sigma_{n-1}^{s_{i,0},u_n}}.\end{aligned} \tag{A47}$$

Finally, to compute the updates of the conditional (co)variances, define

$$\Sigma_{n-1}^{y_n} \equiv \text{Var}\left[y_n \middle| y_1, \ldots, y_{n-1}\right] = \text{Var}\left[y_n - r_{n,n-1}\right], \tag{A48}$$

$$\Sigma_{n-1}^{v,y_n} \equiv \text{Cov}\left[v, y_n \middle| y_1, \ldots, y_{n-1}\right] = \text{Cov}\left[v - p_{n-1}, y_n - r_{n,n-1}\right], \tag{A49}$$

$$\Sigma_{n-1}^{s_{i,0},y_n} \equiv \text{Cov}\left[s_{i,0}, y_n \middle| y_1, \ldots, y_{n-1}\right] = \text{Cov}\left[s_{i,n-1}, y_n - r_{n,n-1}\right], \tag{A50}$$

$$\Sigma_{n-1}^{u_{n+1},y_n} \equiv \text{Cov}\left[u_{n+1}, y_n \middle| y_1, \ldots, y_{n-1}\right] = \text{Cov}\left[u_{n+1} - r_{n,n-1}, y_n - r_{n,n-1}\right]. \tag{A51}$$

In addition, consider the unconditional covariance matrix $\boldsymbol{\Sigma_0}$ in (1) and define for all $n \in \{1, \ldots, N-1\}$ the corresponding conditional covariance matrix $\boldsymbol{\Sigma_n}$ as

$$\boldsymbol{\Sigma_n} \equiv \begin{pmatrix} \underset{(1\times1)}{\Sigma_n^v} & \underset{(1\times M)}{\Sigma_n^{v,\boldsymbol{s}}} & \underset{(1\times N)}{\Sigma_n^{v,\boldsymbol{u}}} \\ \underset{(M\times1)}{\Sigma_n^{\boldsymbol{s},v}} & \underset{(M\times M)}{\Sigma_n^{\boldsymbol{s}}} & \underset{(M\times N)}{\Sigma_n^{\boldsymbol{s},\boldsymbol{u}}} \\ \underset{(N\times1)}{\Sigma_n^{\boldsymbol{u},v}} & \underset{(N\times M)}{\Sigma_n^{\boldsymbol{u},\boldsymbol{s}}} & \underset{(N\times N)}{\Sigma_n^{\boldsymbol{u}}} \end{pmatrix}, \tag{A52}$$

where the (co)variances are conditional on the aggregate order flow history $(y_1, \ldots, y_n)$. Then the property of multivariate Gaussian distributions implies that the conditional

[15] By definition, $\theta_N = 0$ since there are no uninformed orders after the final trading period.

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covariance matrix $\boldsymbol{\Sigma_n}$ can be expressed recursively as

$$\boldsymbol{\Sigma_n} = \boldsymbol{\Sigma_{n-1}} - \begin{pmatrix} \Sigma_{n-1}^{v,y_n} \\ \Sigma_{n-1}^{s_{1,0},y_n} \\ \vdots \\ \Sigma_{n-1}^{s_{M,0},y_n} \\ \Sigma_{n-1}^{u_1,y_n} \\ \vdots \\ \Sigma_{n-1}^{u_N,y_n} \end{pmatrix} \left(\Sigma_{n-1}^{y_n}\right)^{-1} \begin{pmatrix} \Sigma_{n-1}^{v,y_n} \\ \Sigma_{n-1}^{s_{1,0},y_n} \\ \vdots \\ \Sigma_{n-1}^{s_{M,0},y_n} \\ \Sigma_{n-1}^{u_1,y_n} \\ \vdots \\ \Sigma_{n-1}^{u_N,y_n} \end{pmatrix}', \tag{A53}$$

where the (co)variances in Proposition 1 are obtained after simplification. □

G. Algorithm

The equilibrium in Proposition 1 is solved via backward induction. Specify the number of informed traders M, the number of trading periods N, the desired period zero (co)variances Σ_0^v, $\Sigma_0^{v,s_{i,0}}$, Σ_0^{v,u_n}, $\Sigma_0^{s_{i,0},u_n}$, $\Sigma_0^{s_{i,0}}$, $\Sigma_0^{u_n}$, and the constants χ_s and χ_u for computing $\Sigma_0^{s_{i,0},s_{j,0}}$ and $\Sigma_0^{u_n,u_{n+1}}$ from (13) and (14), respectively. Note that the desired specification is only valid if the covariance matrix $\boldsymbol{\Sigma_0}$ in (1) is positive definite.

The algorithm starts in the last trading period N and takes as inputs guesses for the period $N-1$ conditional (co)variances $\Sigma_{N-1}^{v,s_{i,0}}$, Σ_{N-1}^{v,u_N}, $\Sigma_{N-1}^{s_{i,0},u_N}$, $\Sigma_{N-1}^{s_{i,0}}$, and $\Sigma_{N-1}^{u_N}$.[16] Immediately, $\Sigma_{N-1}^{s_{i,0},s_{j,0}}$ follows from (13). By definition, the value function coefficients $a_N, \ldots, k_N$ and the projection coefficient θ_N are zero in the last trading period, simplifying the equations for β_N and λ_N to

$$\beta_N = \frac{\eta_N - \lambda_N \psi_N}{\lambda_N[2 + (M-1)\phi_N]}, \tag{A54}$$

[16] This paper does not compute conditional (co)variances for the last period N since these quantities are realized after the end of the trading game.

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$$\lambda_N = \frac{M\beta_N \Sigma_{N-1}^{v,s_{i,0}} + \Sigma_{N-1}^{v,u_N}}{M\beta_N^2 \left[\Sigma_{N-1}^{s_{i,0}} + (M-1)\Sigma_{N-1}^{s_{i,0},s_{j,0}}\right] + \Sigma_{N-1}^{u_N} + 2M\beta_N \Sigma_{N-1}^{s_{i,0},u_N}}. \tag{A55}$$

Rearranging yields a quadratic equation in λ_N, where only the positive root satisfies the second order condition in period N. Given λ_N, one can compute β_N, ζ_N, γ_N, α_N, δ_N, and the value function coefficients $a_{N-1}, \ldots, k_{N-1}$.

Continuing with the induction step, in every period $n \in \{1, \ldots, N-1\}$ the following quantities are taken as inputs:

(i) The projection coefficient β_{n+1},

(ii) the conditional (co)variances $\Sigma_n^{v,s_{i,0}}$, $\Sigma_n^{v,u_{n+1}}$, $\Sigma_n^{s_{i,0},u_{n+1}}$, $\Sigma_n^{s_{i,0}}$, and $\Sigma_n^{u_{n+1}}$,

(iii) and the value function coefficients $a_n, \ldots, k_n$.

With these inputs one can simultaneously solve for β_n, $\Sigma_{n-1}^{v,s_{i,0}}$, Σ_{n-1}^{v,u_n}, $\Sigma_{n-1}^{s_{i,0},u_n}$, $\Sigma_{n-1}^{s_{i,0}}$, $\Sigma_{n-1}^{u_n}$, resulting in a polynomial system with six equations and six unknowns. Given χ_s and χ_u, one also obtains $\Sigma_{n-1}^{s_{i,0},s_{j,0}}$ and $\Sigma_{n-1}^{u_n,u_{n+1}}$ from (13) and (14), respectively. This in turn allows one to compute λ_n, ζ_n, θ_n, γ_n, α_n, δ_n, and the value function coefficients $a_{n-1}, \ldots, k_{n-1}$ and proceed to the next iteration.

After the final iteration one can compute Σ_n^v for every period $n \in \{1, \ldots, N-1\}$ since Σ_0^v was initially specified. If

(i) the second order condition is satisfied in every period,

(ii) conditional variances are positive and monotonically decreasing over time,

(iii) conditional covariances satisfy the Cauchy-Schwarz inequality in every period,

(iv) and the value function is monotonically decreasing over time,

an equilibrium was found. In addition, if $\Sigma_0^{v,s_{i,0}}$, Σ_0^{v,u_n}, $\Sigma_0^{s_{i,0},u_n}$, $\Sigma_0^{s_{i,0}}$, and $\Sigma_0^{u_n}$ are as desired, the algorithm terminates. Otherwise, the guesses for the period $N-1$ conditional (co)variances are adjusted and the algorithm runs anew.

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